

THE MAJOR PRACTICAL MANUAL OF INVERTEBRATES & CHORDATA OF ZOOLOGY

To prepare such a Major Practical Manual for students and teachers as well as for those people who have scientific attitude and interest for the subject. The current Major practical syllabus of zoology has been framed from Periyar University as well as State level. As there are no such Manual which cover diagrams only but we have insert the original specimen picture were included. We thought to prepare such a Major Practical Manual for students and Professor as well as for those people who have scientific aptitude and interest for the subject and support to prepare semester examination.

The Manual is richly illustrated with figures and original photographs. First aid, precautionary methods and safety rules and regulations and also included specific guidelines, how to handling the equipment in the laboratory.



Dr.S.KALAIMANI is an expert in Biological Research of Zoological Studies in various fields, who currently working as an Assistant Professor of Zoology, P.G and Research Department of Zoology, J.K.K. Nataraja, College of Arts & Science, Kumarapalayam, Namakkal District.. He prepared various simplified Books, Manuals for College students



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Press**

KALAIMANI SUBRAMANI
THANGAM BASKAR
KOWSALYA SHANMIGAM

THE MAJOR PRACTICAL MANUAL OF INVERTEBRATES & CHORDATA OF ZOOLOGY

A syllabus based on Periyar University

**KALAIMANI SUBRAMANI
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Cover image: www.ingimage.com

Publisher:

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Dodo Books Indian Ocean Ltd., member of the OmniScriptum S.R.L
Publishing group

str. A.Russo 15, of. 61, Chisinau-2068, Republic of Moldova Europe

Printed at: see last page

ISBN: 978-613-8-83847-0

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**“THE MAJOR PRACTICAL MANUAL OF
INVERTEBRATES & CHORDATA OF ZOOLOGY”**

(A syllabus based on Periyar University)

FIRST EDITION

JULY-2022

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THE CORE PRACTICAL - I

INVERTEBRATES & CHORDATA OF ZOOLOGY

(A syllabus based on Periyar University)

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By

Mr. S. KALAIMANI, M.Sc., M.Ed., M.Phil, SET (Life Science), SET (E.du), D.C.A.,

ACKNOWLEDGEMENT

Words find no bounds to express my sincere thanks to all who made this practical manual the title of **“The major practical manual of invertebrates & chordata of zoology”** (a syllabus based on periyar university) to come true.

I owe my sincere thanks to Dr.C.Seeranganayaki , Principal, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal D.t, for extending the facilities to undergo Major Practical Manual.

My specific thanks to Dr.S.Umamaheswari, Assistant Professor, PG&Research Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this Major Practical Manual successfully.

I wish to express my profound thanks to All the Staff Members, PG& Research Department of Zoology, J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal Dt.

I also wish to heartfelt thanks to madam Dr.S.Parmeswari, Dean. J.K.K. Nataraja College of Arts and Science, Kumarapalayam, Namakkal D.t for her cooperation and loving encouragement to complete this Major Practical Manual successfully.

My special thanks to Mrs. V.N.Padma, Assistant CTO, for helping me in many ways for completing this Major Practical Manual work. With great pleasure I express my deep sense of feeling of my affectionate Father Late Mr. P.Subramani (late), Sub registrar, my mother Mrs. S.Veerammal (late), my brother Mr. S.Velumani, Junior Asst Registration Department. My children K.Deeptha dhayazhini and K.Mithunmonish.

And last but not least I wish to extend my sincere thanks to All My Friends and Relatives for their co- operation and support in this Major Practical Manual to make success.

Mr. S. KALAIMANI

PREFACE

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The current Major practical syllabus of zoology has been framed from Periyar University as well as State level. As there are no such Manual which cover diagrams only but we have insert the original specimen picture were included.

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Mr. S. KALAIMANI

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UNIT:1 MAJOR PRACTICALS (I.DISSECTION)

A.COCKROACH

1.1.Digestive system

1.2.Nervous system

B.PRAWN

1.3.Nervous system

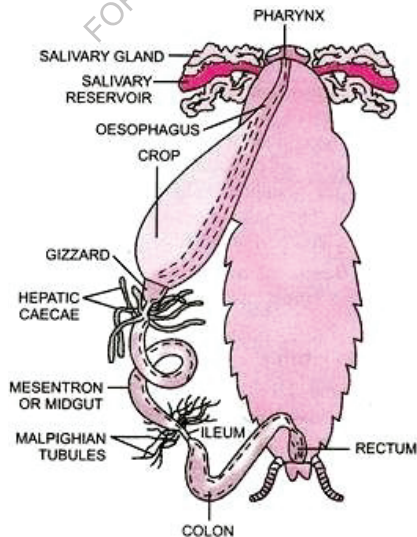
C.ANY BONY FISH

1.4.Digestive system

A. COCKROACH

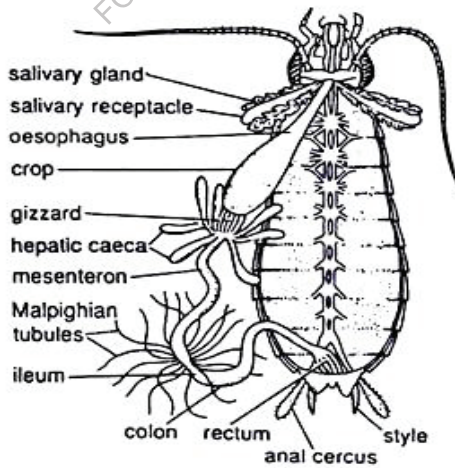
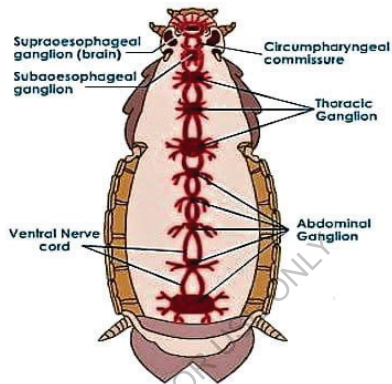
1.1 .DISSECTION PROCEDURE OF DIGESTIVE SYSTEM

1. Hold the specimen with your left hand and clip the wings.
2. Fix the specimen in a dorsal position on a dissecting tray with the help of pins passing through abdominal sterna and coxa of legs.
3. Cut the lateral membrane (pleura) between the terga and sterna of the thorax and abdomen with a pair of fine scissors.
4. Posteriorly the two incisions should meet at the hindmost end of the abdomen. Proceed forward up to the anterior end of the thorax.
5. Give a transverse incision along the anterior border of the first thoracic segment and carefully remove the terga.
6. The thoracic and the abdominal cavity are exposed. Put clear water in the tray. Remove fat bodies and tracheae to expose internal organs.
7. Carefully uncoil the intestine and stretch the alimentary canal to one side .
8. Prevent it from coming back to the original position by pushing down a pin in the wax between the gut and the specimen.



A. COCKROACH

1.2. NERVOUS SYSTEM



A. COCKROACH

1.2. DISSECTION PROCEDURE OF NERVOUS SYSTEM

1. Fix the head of the specimen by pinning through the mandibles.
2. The rest of the body should be fixed to the wax of the dissecting tray in the way already described.
3. Carefully remove the epicranial plate of the head capsule and expose the cerebral ganglia.
4. Cut the pharynx and pull it out with the oesophagus. Remove the viscera and the ventral nerve cord is exposed.
5. Expose the roots of the circumoesophageal connectives on the lateral sides of the brain and trace them to the points where they meet the sub-oesophageal ganglia.

Circumoesophageal connectives:

6. Arising from the brain two short and broad nerves run around the oesophagus to meet the sub-oesophageal ganglia

Sub-oesophageal ganglia:

7. It is formed by the fusion of two ganglia and located in the mid-ventral region of the head just ventral to the oesophagus.
8. Nerves arising from sub-oesophageal ganglia end in labrum, mandible and both pairs of maxillae.
9. A pair of connectives run backward from the ganglia and join the first thoracic ganglion, (prothoracic ganglion).

Ventral nerve cords:

10. These are two solid nerves and run along the mid-ventral line of the thorax and abdomen. The nerve cords are connected by nine ganglia, three thoracic and six abdominal.
11. The three thoracic ganglia and the last abdominal ganglion are large in size.
12. Nerves emanating from the thoracic ganglia innervate the musculature and structures of the thoracic region. Those from the abdominal ganglia send nerves to the structures in the abdominal region.

Stomatogastric or Anterior Sympathetic Nervous System:

13. It consists of a median frontal ganglion, hypo-cerebral ganglion, oesophageal and ingluvial ganglion.

i. Frontal ganglion:

14. Situated in front of brain, on the oesophagus, median in position. Connected to tritocerebral lobes of brain by a pair of frontal connectives.

ii. Hypo-cerebral ganglion:

15. Situated on the oesophagus below the brain. Recurrent nerve connects frontal and hypo-cerebral ganglion.

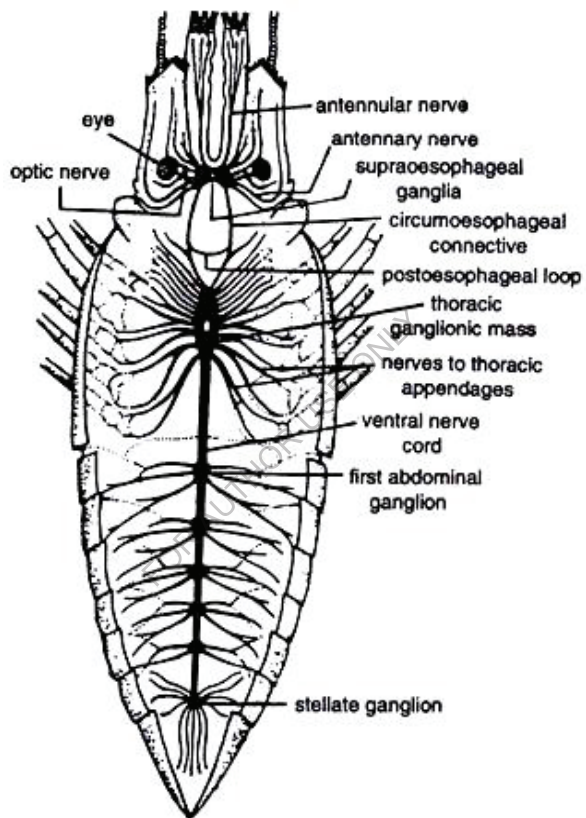
iii. Oesophageal or pharyngeal or occipital ganglion or corpora cardiacum:

16. Situated on both sides of oesophagus slightly above the hypo-cerebral ganglion.
17. From hypo-cerebral ganglia nerves go to corpora cardiaca or oesophageal ganglia.
18. From corpora cardiaca of each side nerve connects ventrally to another ganglia corpora allatum.

iv. Ingluvial or Stomachic ganglion:

19. Situated on the posterior end of the foregut. Nerves from hypo-cerebral ganglion connect the ingluvial or stomachic ganglion.
20. The nerves from ingluvial ganglion innervates the neighbouring regions of the stomodaeum.

B.PRAWN
1.3. NERVOUS SYSTEM



B.PRAWN

DISSECTION PROCEDURE OF NERVOUS SYSTEM

- Remove the carapace and the terga with pleura.
- The brain is located below the base of the rostrum. Remove the chitinous endo-pharyngeal skeletal plates overlying the brain.
- The brain is covered by a thick layer of fat.
- Remove the fat to expose the brain. Remove the hepatopancreas. Cut the oesophagus and pull it out with the stomach.
- Expose the nerves arising from the brain and follow their courses.
- Trace the two circumoesophageal connectives running downwards from the lateral sides of the brain to meet the ventral sub-oesophageal ganglionic mass.
- To expose the ventral nerve cord cut the large flexor muscles of the abdomen along the middle line with a scalpel.
- Press the muscles laterally and pin them down to the wax of the dissecting tray.
- The nerve cord with its ganglia is clearly visible

Brain or supraoesophageal ganglia:

- A bilobed ganglionic paired mass at the base of the rostrum.

Following paired nerves arise from the brain:

- a. Optic supplying the eyes.
- b. Ophthalmic supplying eye muscles.
- c. Antennular supplying antennules.
- d. Antennary supplying antennae.
- e. Tegumental supplying labrum.

Circumoesophageal connectives:

- Arising from the lateral sides of the brain the two nerves run backward and downward round the oesophagus and join the large thoracic ganglionic mass.

Post-oesophageal loop:

- The circumoesophageal connectives are joined by a fine, transverse commissure before they join the thoracic ganglionic mass.

Thoracic or sub-oesophageal ganglionic mass:

Formed by the fusion of 11 segmental ganglia of the cephalothorax and sends 11 pairs of nerves, supplying:

- (1) Mandibles,
- (2) Maxillulae,
- (3) Maxillae, (4-6) 3 pairs of maxillipeds, (7-11) 5 pairs of walking legs (pereopods).

Ventral nerve cord:

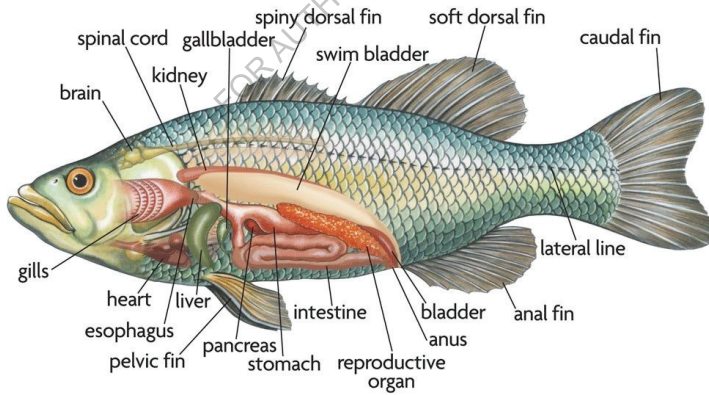
- Runs along the mid-ventral line of the abdomen and bears 6 ganglia.
- The last or the stellate ganglion is the largest. Each ganglion sends nerves to musculature, pleopods and the last one to the uropod.

Visceral Nervous System:

- Two small visceral ganglia, one behind the other; a small nerve connecting the ganglia with the brain and two connectives joining the two visceral ganglia with the circumoesophageal connectives

C. BONY FISH

1.4. DIGESTIVE SYSTEM



C.BONY FISH

1.4. DISSECTION PROCEDURE OF DIGESTIVE SYSTEM

PROCEDURE

1. Obtain a fish, gloves, and dissecting tools.
2. Most of these should already be set out at the lab tables.
3. Look at your fish and orient yourself with the positioning of the fish (e.g. anterior, posterior, dorsal, ventral). Also become familiar with the external structures of the fish.
4. Find the tan-colored liver with the gallbladder attached to and underneath the liver.
5. Cut the liver free from the body to expose the esophagus and stomach.
6. Then, trace the route a piece of food would travel as it passes through the fish's digestive system. Find the mouth, pharynx, esophagus, pyloric caeca (blind ended sacs), stomach, intestines, and anus.
7. To aid in locating other organs, remove the entire digestive canal by cutting it free at the anus and at the mouth.

UNIT:2.MINOR PRACTICALS (MOUNTING)

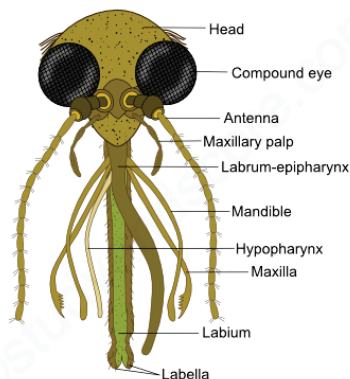
2.1. Mouth parts of House fly, Honey bee, Mosquito

2.2. Sting apparatus of Honey bee

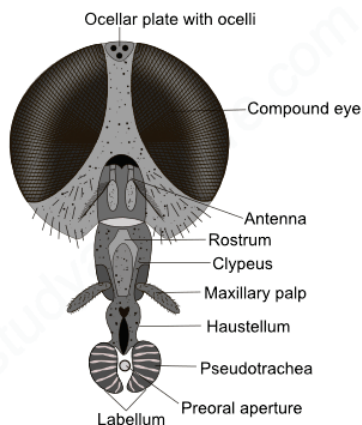
2.3. Scoliodon: Placoid scales

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2.1. MOUTH PARTS OF HOUSE FLY, HONEY BEE, MOSQUITO



MOUThPARTS OF MOSQUITO - PIERCING & CHEWING TYPE

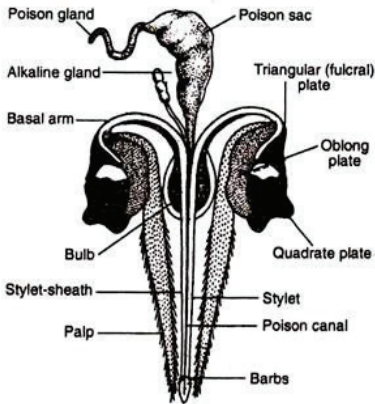


MOUThPARTS OF HOUSE FLY - SPONGING TYPE

PROCEDURE

1. Cut the head from the thorax.
2. Hold the head between the thumb and the index finger with the posterior side of the proboscis facing you in an inverted position.
3. Remove the proboscis by exerting upward pressure at its base with a needle.
4. Place the proboscis in glycerin on a microscope slide and remove the extraneous tissues from it.
5. Place a cover slip over it and examine under a microscope.

2.2. STING APPARATUS OF HONEY BEE

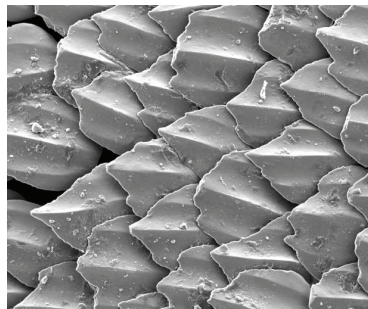
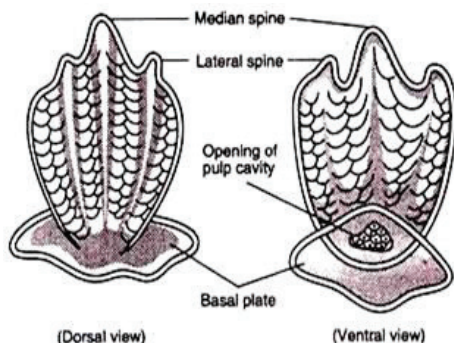


PROCEDURE

1. Provide bees with a source of water and food while housed in the lab. For example, at a minimum, provide access to a 50% (w/v) solution of sucrose in water (this is sufficient for durations of less than 6 hr). For longer periods, provide bees access to honey.
2. House bees in the lab overnight at a temperature of approximately 32 °C and 60-80% relative humidity to reduce stress and avoid dehydration.
3. Using forceps and/or microdissection scissors, remove legs and wings to facilitate dissection of the abdomen. Keep a small beaker or similar container filled with distilled water nearby for the purpose of rinsing instruments between dissections.
4. While restraining the bee with forceps, utilize the microdissection scissors to cut laterally along the dorsal abdominal wall between the first and second tergites
5. While lightly gripping the posterior edge of the second tergite with the forceps, cut longitudinally along each side of the bee from the initial incision to the stinger
6. Use caution when cutting to avoid puncturing the gastrointestinal tract.

2.3. MOUNTING OF PLACOID SCALES OF CARTILAGENOUS FISH

SCOLIODON: PLACOID SCALES



PROCEDURE

1. Take a cartilaginous fish in dissecting tray and cut open the animal with the help of scissor.
2. Remove the skin, cut it into small pieces and put them in petridish.
3. Take 1-2 pieces of skin in a watch glass having water and remove attached muscles with the help of forceps.
4. Transfer the skin pieces into a test tube and add 2-3 ml of 10 % KOH with pipette.
5. Gently boil it for few minutes preferably for 2-3 mins.
6. Allow the test tube to stand for a few minutes. Decant the supernatant leaving a few drops in the test tube.
Pour this residual KOH into a watch glass.
7. Place a drop from it onto a slide and put the cover slip without any air bubble.
8. Observe under low power (10X objective) of a compound microscope.

.A.CLASSIFY GIVING REASON UP TO ORDE

3.1.Paramecium

3.2.Aurelia

3.3.Fasciola

3.4.Ascaris

3.5.Lamellidens

3.6.Asterias

3.7.Balanoglossus

3.8.Herdmania(ascidian)

3.9.Branchiostoma(amphioxus)

3.10.Petromyzon

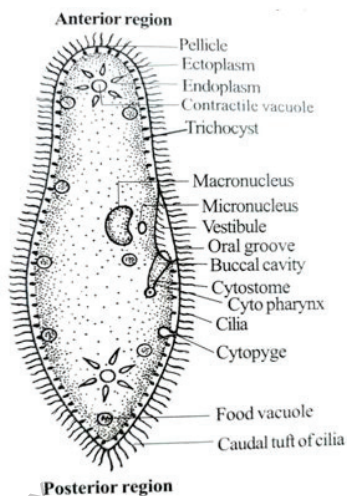
3.11.Scoliodonsorrakowah

3.12.Ranahexadactyla

3.13.Calotesversicolor

3.14.Columbalivia

3.1. PARAMECIUM CAUDATUM



SYSTEMATIC POSITION

Phylum	: Protozoa
Class	: Ciliata
Order	: Holotricha
Family	: Parameciidae
Genus	: <i>Paramecium</i>
Species	: <i>P. Caudatum</i>

PHYLUM: PROTOZOA

1. It is microscopic and acellular.
2. It is free living animal.
3. The body is covered pellicle.

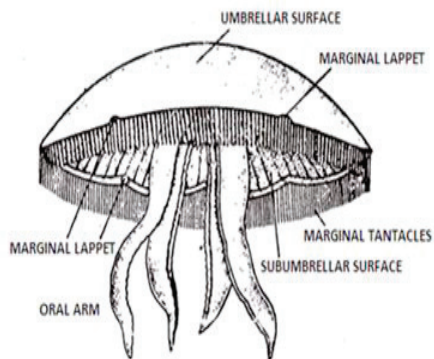
CLASS: CILIATA

1. It moves with the help of cilia.
2. Cilia are persisting throughout the life.

ORDER: HOLOTRICHA

1. The cilia are uniformly arranged in longitudinal rows except at the posterior end of the body.
2. Without an adoral zone of Membranelle and mouth gullet, may or may not present.

3.2. AURELIA AURITA



SYSTEMATIC POSITION:

Phylum	: Cnidaria/ coelenterata
Class	: Scyphozoa
Order	: Semaestomeae
Family	: Ulmaridae
Genus	: <i>Aurelia</i>
Species	: <i>A. aurita</i>

PHYLUM: COELENTERATA

1. Aurelia has radial symmetry with a gastrovascular cavity. Hence it is included in the phylum coelenterate.
2. Diploblastic body wall having outer ectoderm and inner endoderm separated by mesogloea.
3. Respiratory, excretory and circulatory systems are absent.
4. It exhibits the phenomenon of alternation of generation or metagenesis.

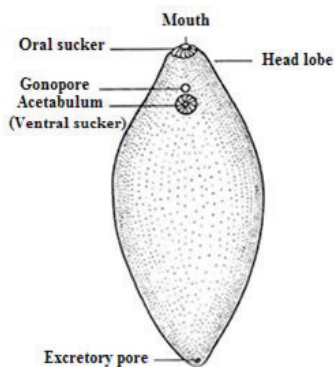
CLASS: SCYPHOZOA

1. Individuals are commonly known as jelly fish.
2. The Aurelia has a true medusa.
3. It is bell or umbrella shaped.
4. Gonads are endodermal origin.
5. Animals are totally marine life.

ORDER: SEMAEOSTOMEAE

1. Umbrella is flat saucer shaped.
2. Craspedic pouches & filaments are present.
3. They are remaining fixed by an aboral stalk to the substratum.
4. Gastro-vascular septa is absent

3.3. LIVER FLUKE (*FASCIOLA HEPATICA*)



SYSTEMATIC POSITION:

Phylum: Platyhelminthes

Class : Trematoda

Order : Digenea

Family : Fasciolidae

Genus : *Fasciola*

Species: *F. hepatica*

PHYLUM: PLATYHELMINTHES

1. It is a acoelomate flattened worm.
2. It is Triploblastic animals

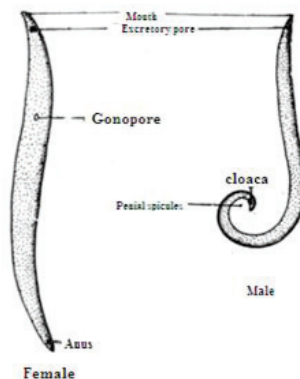
CLASS: TREMATODA

1. All trematodes are parasites.
2. The body is covered by a cuticle.

ORDER: DIGENIA

1. Two suckers are present anterior sucker encircling the mouth and a posterior ventral sucker.
2. Life cycle is complicated with alternation of host

3.4. ASCARIS LUMBRICOIDES



SYSTEMATIC POSITION:

Phylum	: Aschelminthes
Class	: Nematoda
Order	: Ascaridida
Family	: Ascarididae
Genus	: <i>Ascaris</i>
Species	: <i>lumbricoides</i>

PHYLUM: ASHELMINTHES

1. They are pseudocoelomate, triploblastic, bilaterally symmetrical un segmented animal.
2. Body usually worm like cylindrical or flattened.
3. Respiratory and circulatory systems absent.
4. Sexes are separate [dioecious].
5. Cuticle is present.
6. Eggs are shelled, cleavage determinate and spiral.
7. Alimentary canal straight and complete.

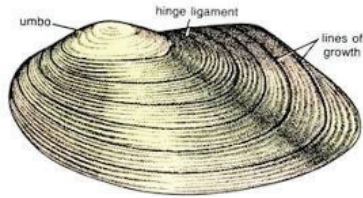
CLASS: NEMATODA

1. They are parasitic elongated roundworm.
2. Body wall consists of cuticle.
3. They are dioeciously.
4. Fertilization is internal.
5. Digestive system complete with muscular pharynx and gland.

ORDER: ASCAROITEA

1. These are parasites of the intestine of vertebrate.
2. Mouth is surrounded by three lips.
3. Pharynx without a posterior bulb.
4. Tail in female is straight and blunt tail in male with two spicule.

3.5.LAMELLIDENS



SYSTEMATIC POSITION:

Phylum	: Mollusca
Class	: Pelecypoda (Bivalvia)
Order	: Eulamellibranchiata
Genus	: Lamellidens
Species	: marginalis

PHYLUM : MOLLUSCA

1. Mollusca are soft bodied animals.
2. They are triploblastic animals.
3. The molluscs are provided with one or two calcareous shells.
4. The shells may be external or internal.

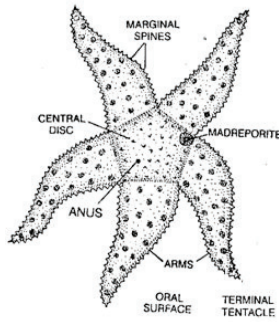
CLASS: PELECYPODA (BIVALVIA)

1. They have bilateral symmetry.
2. The shell of two valves and hence the name Bivalvia.

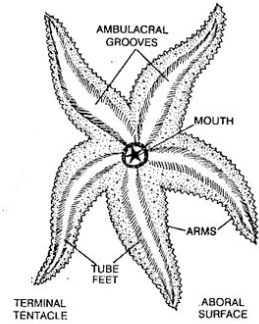
ORDER: EULAMELLIBRANCHIATA

1. Inter filamentary junctions and inter lamellar junctions are vascular.
2. The gills are basket- like.

3.6. STAR FISH (*ASTERIUS RUBENS*)



Dorsal view



Ventral view

SYSTEMATIC POSITION:

Phylum	: Echinodermata
Class	: Asteroidea
Order	: Forcipulatida
Family	: Asteriidae
Genus	: <i>Asterias</i>
Species	: <i>rubens</i>

PHLUM: ECHINODERMATA

1. They are exclusively marine animals.
2. They are radial symmetry. Body is triploblastic.
3. Presence of water vascular system.
4. Nerves system is primitive.
5. True coelom is present.
6. Gills or papulae are the respiratory organs.
7. Reproduction is sexual.

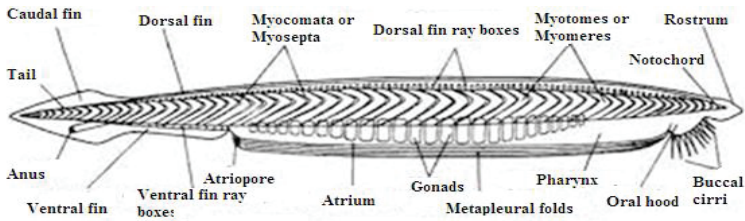
CLASS: ASTEROIDEA

1. The body is pentagonal.
2. Ambulacra grooves open into the tube feet.
3. Respiration by papillae.
4. Oral and aboral surfaces distinct.
5. Development includes bipinnaria or brachiolaria larva.

ORDER; FORCIPULATA

1. Marginal plates inconspicuous or absent.
2. The spines are present on the aboral surfaces.
3. Papulae on both surfaces.
4. Mouth frame of ambulacral.

3.7. AMPHIOXUS (*Branchiostoma lanceolatum*)



SYSTEMATIC POSITION:

Phylum	: Chordata
Subphylum	: Cephalochordata
Class	: Leptocardii
Order	: Amphioxiformes
Genus	: Branchiostoma
Species	: lanceolatum

Phylum: Chordata

Presence of dorsal tubular nerve cord, gill-slits and notochord.

Subphylum: Cephalochordata

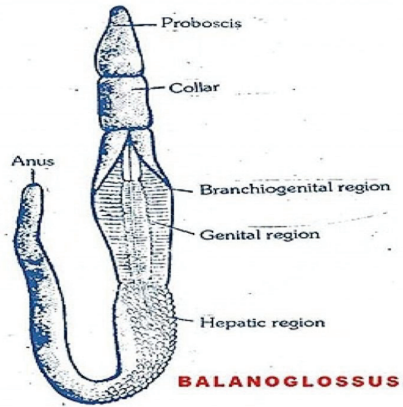
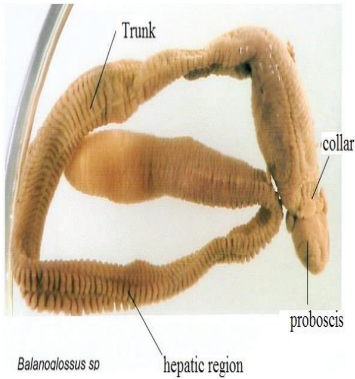
Notochord extends along entire body length.

Class: Leptocardii

Numerous pharyngeal gill-slits present.

1. It is commonly known as lancelet.
2. It is a ciliary feeder.
3. Myotomes are separated by V-shaped partitions called as myocommata.

3.8.BALANOGLOSSUS



SYSTEMATIC POSITION:

Phylum	: Prochordata
Subphylum	: Hemichordata
Class	: Enteropneusta
Order	: Pterobranchia
Genus	: Balanoglossus
Species	: apertus

Phylum: Hemichordata

1. Presence of a short stomochord, notochord in the anterior region. Hence, the phylum hemichordata.

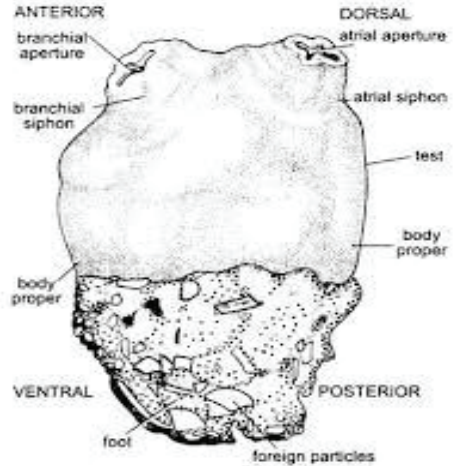
Class: Enteropneusta

2. The body is cylindrical and divided into 3 pairs.
3. It is commonly known as Acorn worm.
4. They have a vermiform body and are found on sandy beaches near seas in warm climate.

Family: Pterobranchia

5. They have a tubular dwelling and live in deep sea-waters.
6. They are bottom dwellers who attach to other organisms for their survival.

3.9. HERDMANIA (*ASCIDIAN*)



SYSTEMATIC POSITION:

Phylum : Prochordata
Subphylum : Urochordata
Class : Ascidiacea
Order : Stolidobranchia
Species : Ascidia

PHYLUM : CHORDATA

1. Notochord is present
2. Dorsal nerve cord and gill slits are present

SUB-PHYLUM : UROCHORDATA

3. It is Marine animal.
4. Body is covered by a thick test.
5. Notochord is present only in larval tail

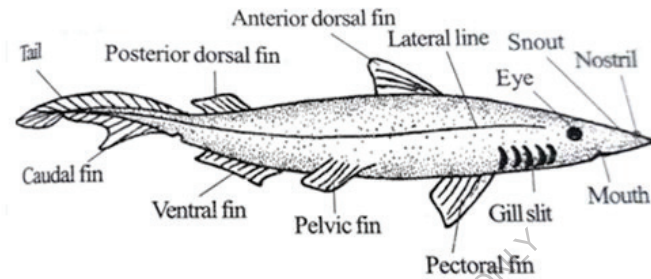
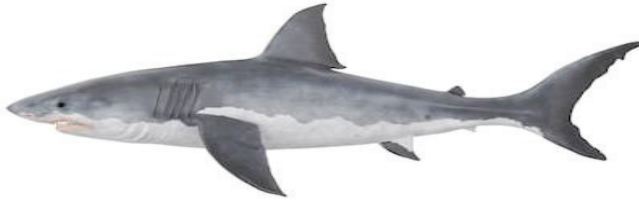
CLASS : ASCIDIACEA

6. It has test with scattered muscles.
7. It has many pharyngeal gill-slits.
8. It lives bottom-dwelling animal.

ORDER: ENTEROGONA

8. It has neural gland ventral to nerve ganglion.
9. It has only one Gonad .

3.10. SHARK (*Scoliodon sorrakowah*)



SYSTEMATIC POSITION:

Phylum: Chordata

1. Dorsal and hollow nerve cord
2. A longitudinal supporting rod-like notochord
3. Pharyngeal gill slit

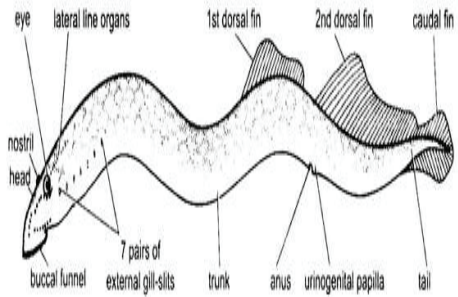
Class: Chondrichthyes

4. Body is covered with minute placoid scales.
5. Endoskeleton cartilaginous.
6. Ventrally placed mouth with rows of teeth.

Order: Lamniformes

7. Both dorsal fins not supported by spines.
8. Anal fin present.
9. Five gill arches.

3.11. PETEROMYZON (LAMPREY)



SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Cyclostomata
Order	: Petromyzontia
Genus	: petromyzon
Species	: marinus

PHYLUM: CHORDATA

1. Notochord is replaced by vertebral column
2. Distinct cranium or brain case is present.
3. Presence of well developed head.

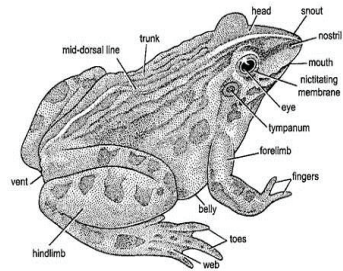
CLASS: CYCLOSTOMATA

4. Mouth is circular and suctorial
5. Naked and glandular skin
6. Continuous and persistent notochord.

ORDER: PETROMYZONTIA

7. Dorsal fin well developed.
8. Well developed branchial basket.
9. Metamorphosis present in the life cycle.
10. External parasites on fish.

3.12. RANA HEXADACTYLA

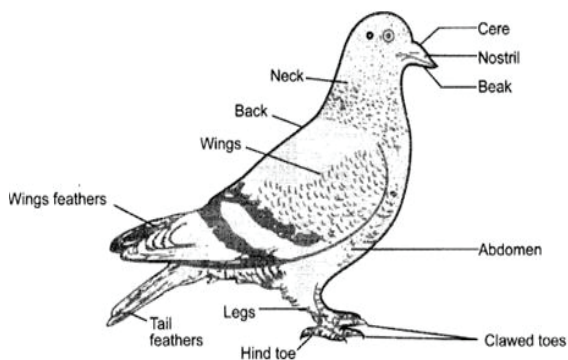


SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Amphibia
Family	: Ranidae
Order	: Anura
Genus	: <i>Rana</i>
Species	: <i>hexadactyla</i>

1. They contain a cranium around the brain.
2. Notochord is replaced by vertebral column.
3. Pharyngeal gill slit
4. It is cold-blooded vertebrates with dual life
5. They have jaws.
6. The limbs are pentadactylous.
7. Order : Anura
 - a. 1. Body short, broad and trunk fused.
 - b. 2. Neck and tail absent
 - c. 3. Fore limbs short, hind limbs long and help in long jump.

3.13. PIGEON (*Columba livia*)



SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Aves
Order	: Columbiformes
Genus	: <i>Columba</i>
Species	: <i>livia</i>

Phylum: Chordata

1. Notochord is replaced by vertebral column
2. Distinct cranium or brain case is present.
3. Presence of well developed head.

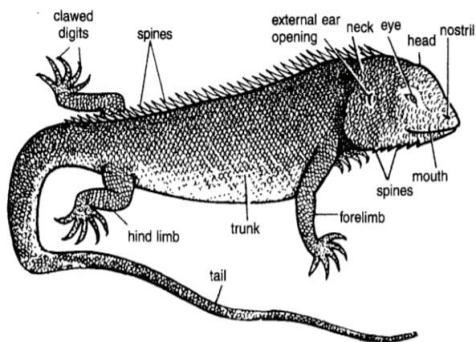
Class : Aves

4. Body is covered with Feathers
5. Head is small, fairly long movable neck and mouth is provided with beaks.

Order : Columbiformes

6. Head and neck are small with short, slender beak.
7. Skin soft and thick, wings long and pointed.

3.14. CALOTES VERSICOLAR



SYSTEMATIC POSITION:

Phylum	: Chordata
Class	: Reptilia
Order	: Squamata
Genus	: <i>Calotes</i>
Species	: <i>versicolor</i>

Phylum: Chordata

1. Notochord is replaced by vertebral column
2. Distinct cranium or brain case is present.
3. Presence of well developed head.

Class: Reptilia

4. Skin dry, cornified, usually covered with epidermal scales or scutes.
5. A post anal tail is usually present
6. well developed eyes, provide with eyelids and nictitating membrane.

Order: Squamata

7. Body is covered by epidermal scales
8. Teeth are pleurodont.
9. Quadrate is movable.
10. Cloacal opening is transverse

B.DRAW LABELLED SKETCHES

3.15.Obelia medusa

3.17.Ephyra larva

3.18.Redia larva

3.19.Cercaria larva

3.20.Mysis larva

3.21.Alima larva

3.22.Bipinnaria larva

3.23.Amphioxus-T.S.through pharynx

3.24.Doliolum

3.25.Salpa

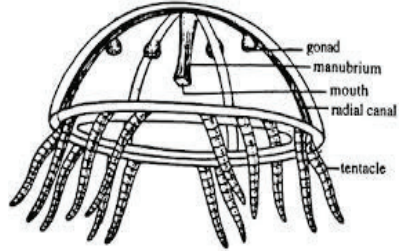
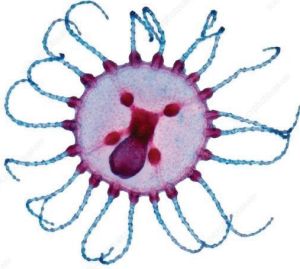
3.26.Narcine

3.27.Cynoglossus

3.28.Alcedoatthis(king fisher)

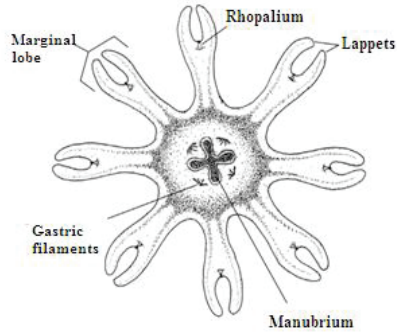
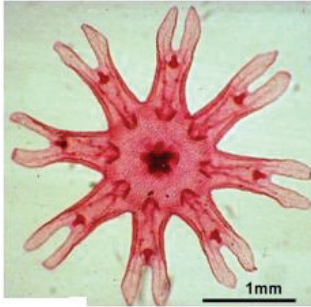
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3.16 .OBELIA MEDUSA



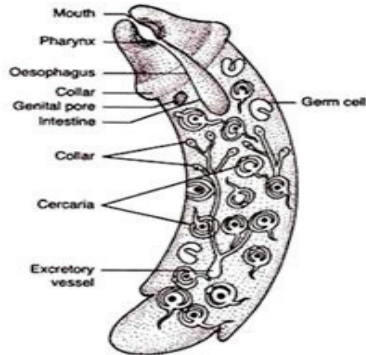
1. Obelia is a marine animal that belongs to the phylum Cnidaria.
2. It has two reproductive stages in its life cycle known as polyps and medusa.
3. Polyps are the asexual reproductive phase whereas medusa is the sexual reproductive phase.
4. Obelia is a very tiny marine hydroid with a height of 2 cm or a little more.
5. The life history of obelia exhibits alternation of generations (between sexual and asexual) and is termed as metagenesis.
6. The outer surface of medusa is known as the ex-umbrellar surface and is convex in shape.
7. The inner surface of the medusa is known as the sub-umbrellar surface and is concave in shape.
8. There is a short manubrium that has a quadrangular mouth at its distal end. This manubrium hangs from the center of the inner concave surface.
9. Initially, the margins of a medusa have 16 contractile short tentacles. These tentacles gradually increase in number.
10. Medusae are dioecious, i.e., male and female medusae are different individuals.

3.17. EPHYRA LARVA OF *AURELIA*



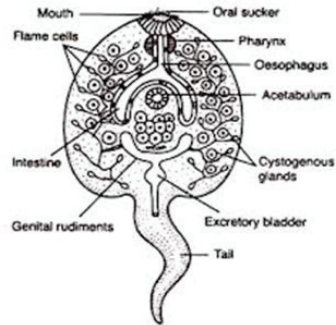
1. Ephyra larva develops from scyphistoma by strobilation
2. It is a free swimming Larva
3. It is a medusoid form.
4. It is umbrella-shaped. It has eight bilobed arms.
5. The gastrovascular system with gastric filaments is present.
6. The larva swims actively and feeds on minute organisms
7. After metamorphosis it develops into Aurelia
8. Thus, the stages in the life history of Aurelia are male and female medusae → gametes → zygotes → ciliated planula larvae → hydratubae → scyphistomae → ephyra → medusa.
9. In Aurelia Aurita, there are several phases that show variations which are found in the life-cycle of other Scyphozoa. Very large eggs of Aurelia become actinula larvae (an advanced type of planula which is like a polyp and has a short stem) which directly form ephyrae larvae.
10. Smaller eggs form planula larvae which become hydratubae and then scyphistomae which strobilate ephyrae.
11. In Aurelia if the food is plentiful and the temperature is low, then many ephyrae are produced at one time (polydisc strobilation).
12. If there is less food and the temperature is high then ephyrae are produced one at a time (monodisc strobilation).

3.18. REDIA LARVA OF LIVER FLUKE



1. Redia larva develops from the germ cells of sporocyst.
2. It lives in the digestive glands of the snail.
3. It is cylinder shaped. The body is covered by cuticle.
4. Behind the mouth, there is a muscular ring called collar.
5. Near the posterior end a pair of projections is found.
6. The mouth leads into a pharynx which ends in a sac-like intestine.
7. The pharynx is surrounded by a group of pharyngeal glands.
8. Two protonephridia are located inside the body. Each protonephridium is formed of many flame cells. It opens to the outside by nephridiopore.
9. The cavity of the larva is filled with germ cells. The germ cells of redia develop into daughter redia.
10. The germ cells of daughter redia develop into the next larva called cercaria. They come out through the birth pore.

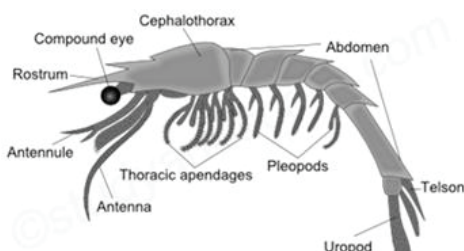
3.19. CERCARIA LARVA OF LIVER FLUKE



COMMENTS:

1. Cercaria (plural cercariae) is the larval form of the trematode class of parasites.
2. It develops within the germinal cells of the sporocyst or redia.
3. A cercaria has a tapering head with large penetration glands.
4. It may or may not have a long swimming "tail", depending on the species.
5. The motile cercaria finds and settles in a host where it will become either an adult, or a mesocercaria, or a metacercaria, according to species.
6. Rotifers (*Rotaria rotatoria*) produce a chemical, Schistosome Paralysis Factor, suppressing cercaria swimming and reducing infections.

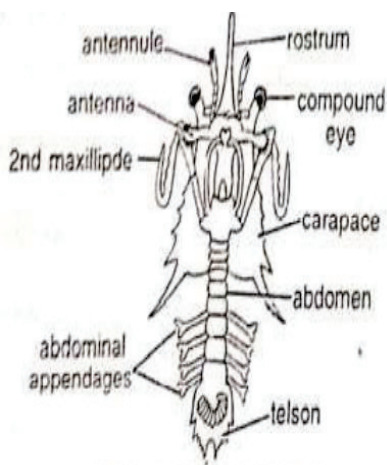
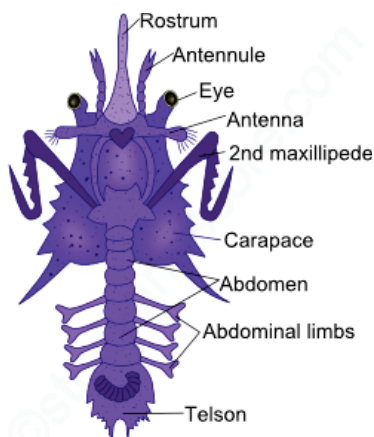
3.20. MYSIS LARVA OF *PENAEUS*



COMMENTS:

1. In *Penaeus*, the zoea larva, instead of converting into the megalopa stage, moults into the post larval mysis larva.
2. It is elongated, laterally compressed and transparent.
3. The body is divisible into cephalothorax and abdomen.
4. The cephalothorax is covered by a carapace.
5. The appendages are well developed. They include 5 pairs of cephalic appendages, 8 pairs of thoracic appendages and 6 pairs of abdominal appendages.
6. The thoracic appendages are used for swimming.
7. The eyes are stalked and compound.
8. The abdomen develops similar to that of the adult form, with 5 pairs of biramous pleopods and a pair of uropods and a telson.
9. The mysis larva metamorphosis into the adult prawn by the loss of the exopodites on the thoracic legs.
10. The larval mysis differs from the adult mysis in the absence of a brood pouch beneath the thorax and a pair of statocysts.

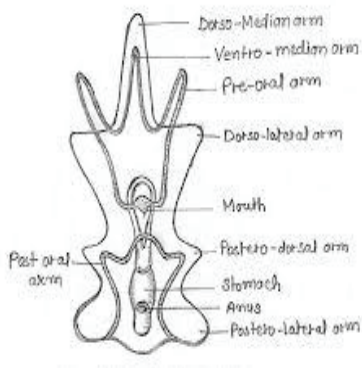
3.21. ALIMA LARVA



COMMENTS:

1. The Alima larva of Squilla hatches out from the egg directly.
2. It is a modified zoaea larva form.
3. It is a pelagic larva, having a glass-like transparency and occurring in large numbers in the plankton. It has a slender form, and a short and broad carapace.
4. All the head appendages are present. But only is 6-segmented, having 4 or 5 pairs of pleopods.
5. The Alima larva differs from the zoaea larva in the armature of the telson and a very large raptorial second maxillipedes.
6. Alima larva is modified form of zoaea found in some malacostracan (e.g. Squilla) which hatches from egg.
7. It is a pelagic form with glassy transparency having a slender body. It bears short and broad carapace.
8. It has all the cephalic appendages but only first two thoracic ones.
9. In segmented abdomen with 4 or 5 pairs of pleopods, is present. It differs from zoaea in having well formed second maxillipedes and the armature of the telson

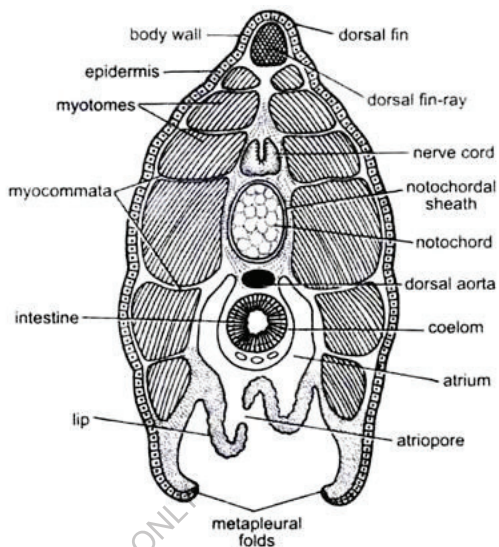
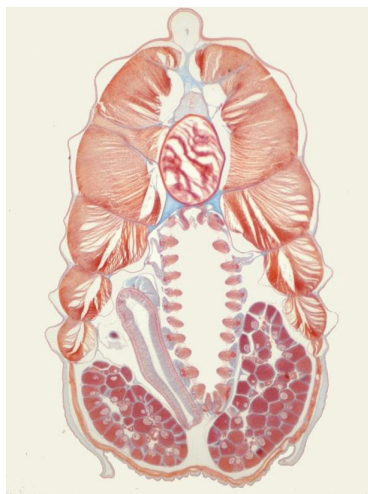
3.22. BIPINNARIA LARVA



COMMENTS:

1. Bipinnaria is the second larva of starfish. It develops from dipleurula larva.
2. After Gastrulation the egg hatches into a larva which develops cilia and begins a free-swimming life
3. It is minute and microscopic.
4. It swims freely on the surface of the water.
5. It is bilaterally symmetrical.
6. At the anterior end dorso-median arm, and pre-oral arms are present.
7. On lateral sides antero- dorsal arms, postero- dorsal arms, post-oral arms and postero- lateral arms are present. Alimentary canal consists of month, oesophagus, stomach, intestine and anus.

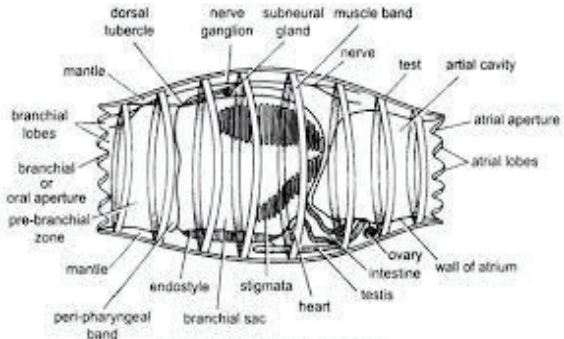
3.23. AMPHIOXUS–T.s.through pharnx



COMMENTS:

1. Body wall composed of epidermis, dermis and muscle layer.
2. Mycomma divide muscle layer into myomees.
3. Presence of dorsal fin supported by fin ray box. 3. Presence of section of nerve cord, notochord, orali cirri.
4. Root of the buccal cavity bear a hatschek's pit.
5. Presence of hatschek's nephridium.
6. Presence of oral or buccal caity.
7. Oral hood supported by skeletal rods.

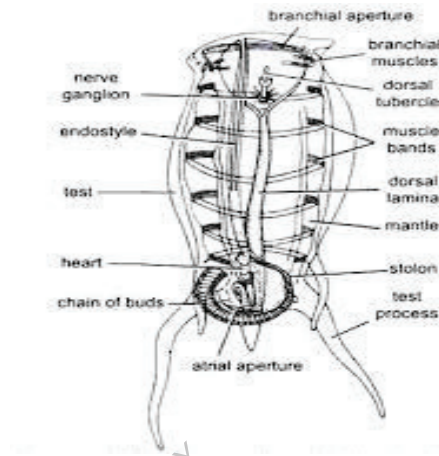
3.24. DOLIOLUM



COMMENTS:

1. The Doliolidium is a small marine animal. It is free-swimming, pelagic form inhabiting the warm water.
2. The doliolid body is small, typically 1–2 mm long.
3. Body: barrel-shaped with the mouth and at opposite end lies anus.
4. Test or tunic, thin, transparent and has no cells.
5. Open ends of barrel are fringed with lobes.
6. In voluminous pharynx oral funnel opens broadly.
7. Pharynx contains perforated by dorsal and rows of stigmata present ventrally
8. In the pharynx Endistyle and peripharyngeal ciliated bands are present
9. Absence of dorsal lamina and tentacles
10. It has small Stomach, short intestine.
11. It is Hermaphrodite
12. Near the anterior end dorsal ganglion and subneural glands are situated.
13. Doliolids alternate through sexual and asexual generations

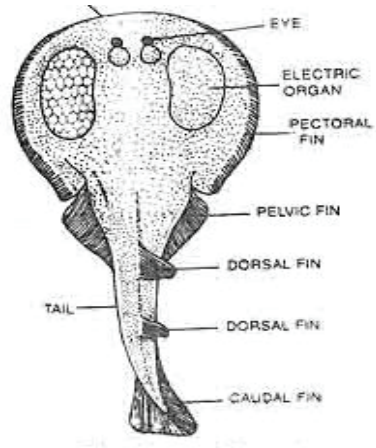
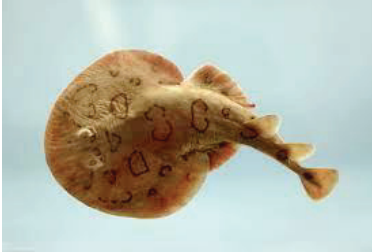
3.25. SALPHA



COMMENTS:

1. Salpa is free swimming and solitary pelagic forms
2. Body of Salpa has barrel-shaped with the mouth and atrial aperture present at opposite end
3. It has transparent test
4. It has Endostyle and pharyngeal bands
5. Dorsal lamina is called as gills.
6. On dorsal surface a single tentacle is present
7. Digestive system and heart is present in a spot called nucleus
8. Eye is present near the nerve ganglion
9. In front of the pharynx dorsal tubercle is present
10. Presence of subneural gland ventral to the ganglion
11. Salpa is found in 2 forms:
 - a) Asexual phase or oozoid
 - b) Sexual phase or blastozoid.

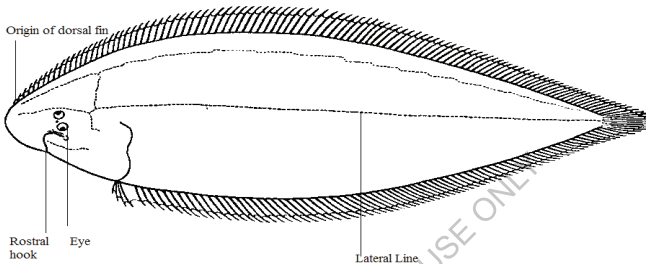
3.26. NARCINE



COMMENTS:

1. Narcine is commonly called electric ray. It is a cartilaginous fish. It is marine.
2. This body consists of a head, a trunk and a tail.
3. The 'head' is disc-shaped and dorso ventrally flattened.
4. The pectoral fin is extended toward to form the disc.
5. Skin is smooth without scales. Mouth is ventral.
6. Eyes are dorsal. Two spiracles are present behind the eyes. Gill slits are ventral.
7. The trunk has a pair of pelvic fins and two dorsal fins.
8. The caudal fin is rounded
9. Narcine has two large electric organs on the disc.
10. They are kidney-shaped.
11. Each electric organ is formed of muscle fibers arranged in blocks and serve as batteries.
12. The electric organs are used for offense and defense.

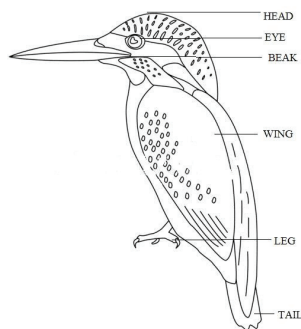
3.27. Cynoglossus



COMMENTS:

1. Lance- or tongue-shaped flatfishes with eyes on left side of body (size to 40 cm).
2. The body highly compressed and tapering to a point posterior.
3. Eyes small and usually close together. Mouth small, sub terminal, asymmetrical; reaching posteriorly to point between verticals at anterior and posterior margins of lower eye or slightly posterior to lower eye; jaws mod
4. Teeth minute and usually only on blind-side jaws; lips fringed with labial papillae in Paraplagusia.
5. Rostral hook usually present below mouth. Posterior margin of preopercle strongly attached to opercle, without free margin and covered with skin and scales.
6. No spiny rays in dorsal, anal, and pelvic fins; dorsal fin reaching far forward onto head; dorsal and anal fins confluent with caudal fin; pectoral fins absent; usually only right pelvic fin present.
7. Scales small, ctenoid or cycloid (smooth).
8. Lateral lines variable; none present in Symphurus; 1 to 3 on eyed side and 0 to 2 on blind side in Cynoglossus and Paraplagusia.
9. Species of Symphurus with only a single proximal dorsal-fin pterygiophore inserted into first interneural space; species of Cynoglossus and Paraplagusia with several pterygiophores inserted into first interneural space

3.28. ALCEDOATTHIS (KING-FISHER)



COMMENTS:

1. Most species are tropical in distribution, and a slight majority are found only in forests.
2. The Kingfisher is a family of medium-sized birds known for their fish-hunting feeding behavior.
3. In general, all kingfishers are medium-sized birds with long pointy bills that aid in capturing fish and other prey.
4. Most species have bright plumage, often bright green, blue and orange hues such as those seen on the common European kingfisher.
5. Most also have long, sharp bills. Many kingfishers have short legs with four toes on each foot with three pointing forward.
6. They have dark brown eyes that provide excellent vision, allowing them to spot their prey from a great distance before attacking.
7. All kingfishers will form nests, often in tree cavities or holes dug into the banks of a river,
8. Kingfishers feed on a wide variety of prey.
9. They are most famous for hunting and eating fish.
10. They are generally monogamous.
11. Most species have bright plumage with only small differences between the sexes.

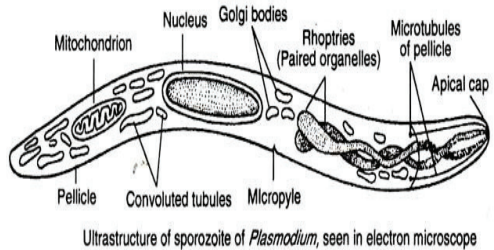
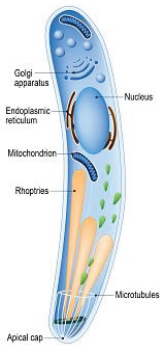
C.COMMENT ON BIOLOGICAL SIGNIFICATION

- 4.1. Plasmodium
- 4.2. Obelia colony
- 4.3. Physalia
- 4.4. Vellela
- 4.5. Fasciola-miracidium
- 4.6. Teania-mature proglottid
- 4.7. Trochophore larva
- 4.8. Chaetopterus
- 4.9. Peripatus
- 4.10. Hirudinaria
- 4.11. Limulus
- 4.12. Chiton
- 4.13. Sepia
- 4.14. Octopus
- 4.15. Tornaria larva
- 4.16. Ascidian tadpole larva
- 4.17. Anabasscandens
- 4.18. Clariasbatrachus
- 4.19. Hippocampus
- 4.20. Echeneis
- 4.21. Ichthyophis
- 4.22. Axolotle larva
- 4.23. Chamaeleon
- 4.24. Viperarusselli(Russell viper)
- 4.25. Dracovolans
- 4.26. Dinopium(wood pecker)
- 4.27. Bat

FOR AUTHOR USE ONLY

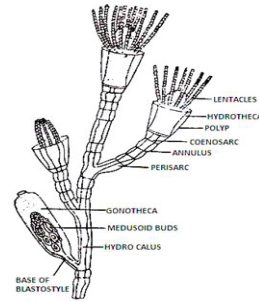
4.1. BIOLOGICAL SIGNIFICANCE OF PLASMODIUM

Plasmodium



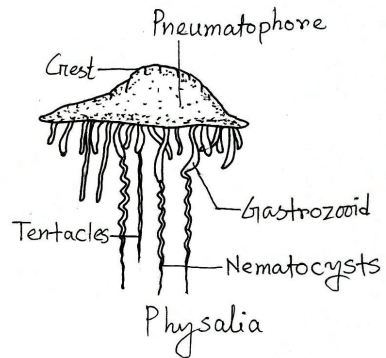
- Malaria is transmitted primarily by the bite of infected Anophelines mosquitoes. It can also be transmitted by inoculation of infected blood and congenitally.
- Anophelines feed at night and their breeding sites are primarily in rural areas. The greatest risk of malaria is therefore from dusk to dawn in rural areas.
- In many malaria-endemic areas, there is little or no risk in urban areas. However, urban transmission is common in some parts of the world, especially Africa.
- Malaria continues to be the most important tropical disease affecting humans. The condition is caused by protozoa of the genus *Plasmodium*. Infection is transmitted to humans by the female Anophelines mosquito.
- The genus *Plasmodium* includes > 170 different species that infect mammals, reptiles, birds, and amphibians. Four species have long been known to cause malaria in humans: *Plasmodium falciparum*, *P. vivax*, *P. ovale*, and *P. malariae*.
- More recently, *P. knowlesi*, which normally infects long-tailed and pig-tailed macaque monkeys, has been implicated as a cause of human malaria in South East Asia, Borneo, Thailand, Singapore, and parts of the Philippines.
- The disease is caused by one of the five different *Plasmodium* species (see above), but the majority of the morbidity and mortality attributed to malaria, especially central nervous system (CNS) involvement, are caused by *P. falciparum*.
- The exception of the Caribbean and parts of Central America, *P. falciparum* strains are usually resistant to chloroquine.

4.2 .BIOLOGICAL SIGNIFICANCE OBELIA



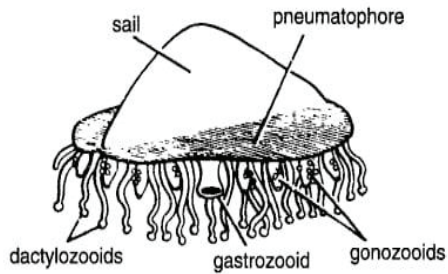
- Obelia is used in many zoology textbooks to exemplify the Hydrozoa life cycle.
- The body plan of Obelia medusae, however, is unique among the Hydrozoa. Contrary to all other hydromedusae, Obelia medusae do not derive from a medusary nodule, have neither a velum nor a sub umbrellar cavity, the umbrella is flat, and swimming is achieved by umbrellar flapping and does not involve jet propulsion.
- Medusae of most Obelia species do not grow well on a diet of *Artemia salina* nauplii, the usual food for laboratory-reared hydrozoans, and their rearing is often difficult.
- Previous observations reported that Obelia medusae might eat phytoplankton, driven to the mouth by movements of both the bell and the tentacles.
- To evaluate the efficiency of this feeding mechanism, we gave a suspension of genetically modified *Escherichia coli* expressing GFP to starved, newly released medusae of both *O. dichotoma* and *Clytia hemisphaerica* (the other campanulariid genus with medusae).
- Obelia medusae are able to concentrate bacteria in their gastric cavity, whereas *Clytia* medusae are not. Contrary to other Hydromedusae that are macrophagous,
- Obelia is a microphagous and filter-feeding medusa, at least at the onset of its medusan life. As Cnidaria evolved before their current metazoan prey, their ancestral diet was probably microphagous.
- Obelia's microphagous feeding habits could thus represent an example of the ancestral feeding condition in Cnidaria, but the apomorphic characters of its medusa suggest that its body architecture is not ancestral, and that it's microphagy evolved relatively recently.

4.2. BIOLOGICAL SIGNIFICANCE PHYSALIA



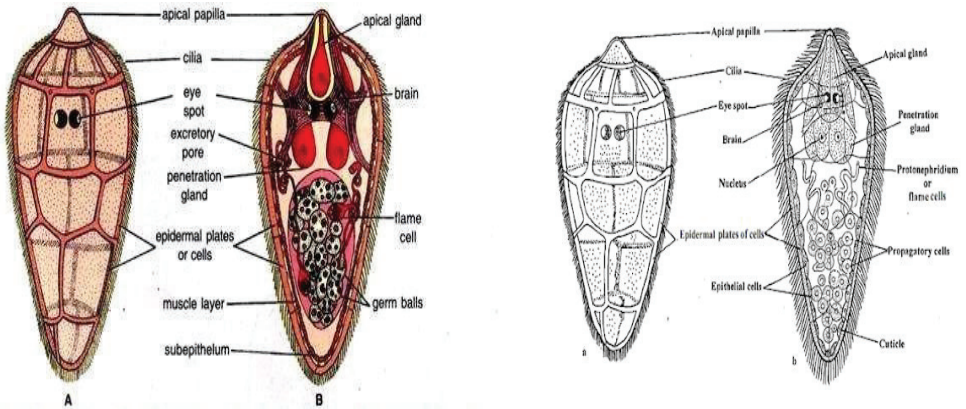
- The Portuguese Man-of-War is eaten by some fish and crustaceans (e.g. the sand crab) that can be of commercial value.
- This species can hurt tourists and tourism in areas where it is common, due to stings (of neurotoxins) from its Cnidocytes.
- Much money is spent each year to treat swimmers who have been stung by the tentacles of individuals that have washed up on beaches.
- The inflammatory response resulting from stings is due to the release of histamines from mast cells within the victim.
- Injures humans
- Bites or stings
- Venomous

4.3. BIOLOGICAL SIGNIFICANCE VELLELA



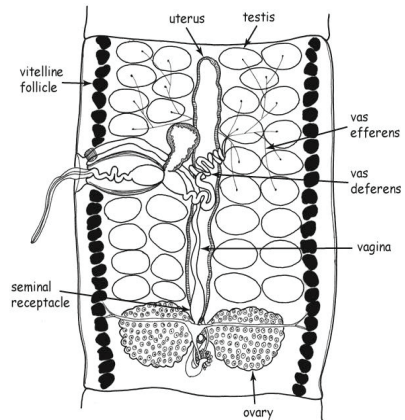
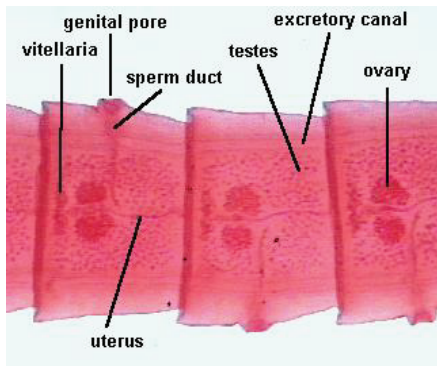
- Velella is a monospecific genus of hydrozoa in the Porpitidae family. Its only known species is Velella velella,
- A cosmopolitan free-floating hydrozoan that lives on the surface of the open ocean. It is commonly known by the names sea raft, by-the-wind sailor, purple sail, little sail, or simply Velella.
- This small cnidarian is part of a specialised ocean surface community that includes the better-known cnidarian siphonophore, the Portuguese man o' war. Specialized predatory gastropod molluscs prey on these cnidarians.
- Such predators include nudibranchs (sea slugs) in the genus Glaucus and purple snails in the genus Janthina.
- Like other Cnidaria, Velella velella are carnivorous. They catch their prey, generally plankton, by means of tentacles that hang down in the water and bear cnidocysts (also called nematocysts).
- V. velella's nematocysts are relatively benign to humans, although each person may respond differently to contact with the nematocyst toxin. It is wise to avoid touching one's face or eyes after handling V. velella, and itching may develop on parts of the skin that have been exposed to V. velella nematocysts.

9.5. BIOLOGICAL SIGNIFICANCE FASCIOLA –MIRACIDIUM



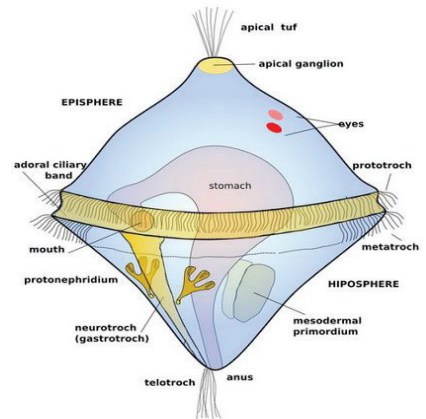
- Liver flukes cause tremendous loss to farmers of cattle and sheep.
- They are responsible for such diseases as liver rot and black disease, which are detrimental to livestock.
- The first larval stage, the miracidium, generally is free-swimming and penetrates a freshwater or marine snail, unless it has already been ingested by one.
- Within this intermediate host, the parasite passes through a series of further stages known as sporocysts, rediae, and cercariae.

9.6. BIOLOGICAL SIGNIFICANCE TAENIA MATURE- PROGLOTTID



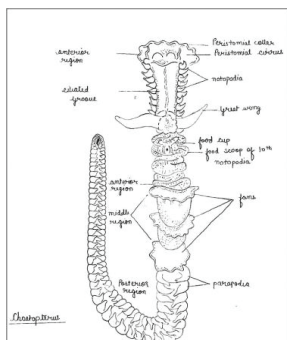
- Although some platyhelminths (flatworms) are free-living and nondestructive, many other species (particularly the flukes and tapeworms) parasitize humans, domestic animals, or both.
- In Europe, Australia, and North and South America, tapeworm infestations of humans have been greatly reduced as a consequence of routine meat inspection.
- But where sanitation is poor and meat eaten undercooked, the incidence of tapeworm infestations is high.
- In the Baltic countries much of the population is infested with the broad tapeworm (*Diphyllobothrium latum*); in parts of the southern United States a small proportion of the population may be infested with the dwarf tapeworm (*Hymenolepis nana*).
- In Europe and the United States the beef tapeworm (*Taenia saginata*) is common because of the habit of eating undercooked steaks or other beef products.

9.7. BIOLOGICAL SIGNIFICANCE TROCHOPHORE LARVA



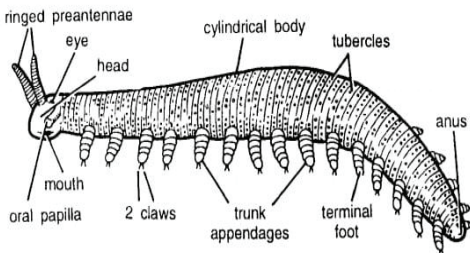
- Trochophore, also called trochosphere, small, translucent, free-swimming larva characteristic of marine annelids and most groups of mollusks.
- Trochophores are spherical or pear-shaped and are girdled by a ring of cilia (minute hairlike structures), the prototroch, that enables them to swim.
- When development is indirect, there is a larval stage which differs from the adulthood through metamorphosis. Advancement from free living to sessile or sedimentary requires many stages.
- If the food habit scenario limits the larval form and adults are coexisting member of a habitat, more taxa shows a significant larva. As in case phylogeny points out their origin to be a monophyletic one.
- Such as larva is Trochophore arising theories in bilateral over radial symmetry.
- Etymology: Trochophore derives from the ancient greek where trokhos meaning "wheel", and phero means 'to bear, to carry', because the larva is bearing a wheelshaped band of cilia.
- Characteristics larval stage of annelids, egg hatch to form Trochophore larva- the first larval stage.
- Also found in Mollusca and some other phyla also.

9.8. BIOLOGICAL SIGNIFICANCE CHAETOPTERUS



- Chaetopterus or the parchment tube worm or parchment tube worm is a genus of marine polychaete worm that lives in a tube it constructs in sediments or attaches to a rocky or coral reef substrate.
- They are planktonic in their juvenile forms, as is typical for polychaete annelids.
- Species include the recently discovered deep water Chaetopterus pugaporcinus and the well-studied Chaetopterus variopedatus.
- Members of the genus have no organs for detecting light and live their entire adult lives in their buried tubes. In spite of this, known members of the genera exhibit strong bioluminescence.
- species of filter-feeding crustaceans from different families may make their homes in the Chaetopterus tube, Polyonyx gibbesi (family Porcellanidae) and Pinnixa chaetoptera (family Pinnotheridae)
- Polyonyx and Pinnixa are almost always present in the tubes, generally as a pair of crabs, and both species may inhabit the tubes at the same time.
- The crustaceans may live most of their lives in the tubes, leaving in the case of disturbance by exiting through the chimneys at either ends or, when the chimneys are too narrow, by biting through the parchment tube.
- It is hypothesised that this nudibranch feeds on the mucus net which Chaetopterus uses to trap its food or on the faeces of Chaetopterus.
- Tenellia rolleri is another nudibranch with similar flattened morphology which lives in muddy areas with no obvious food source, but abundant Chaetopterus tubes.
- It is possible that it lives and feeds in a similar manner.

4.9. BIOLOGICAL SIGNIFICANCE OF PERIPATUS



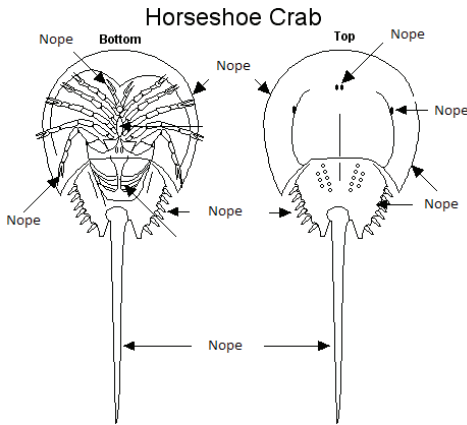
- It is presumed that Onychophora represents a phylogenetically intermediate stage between Annelida and Arthropoda. However, some of their special adaptations do not confirm the “missing link” status of Onychophora.
- Peripatus has no economic importance but zoologically it is very interesting because it possess the characters of both Annelida and Arthropoda as well as the peculiarities of its own.
- Onychophora has the characters of both Phylum Annelida and Phylum Arthropoda. Therefore, they are regarded as the intermediate stage or the connecting link between Annelida and Arthropoda.
- However, they appear to be more closely allied to arthropods than to annelids and have arisen as an offshoot from the base of arthropod line.
- In fact onychophorans are neither worms nor arthropods but have distinct characters of their own.
- Its extreme isolation and discontinuous distribution at present suggest that this group has lived through many geological ages and that it has been more wide spread and diversified in past.
- Aysheaia and Peripatus suggest that arthropoda may have evolved from annelid like ancestors.

4.10. BIOLOGICAL SIGNIFICANCE HIRUDINNARIA



- *Leeches* are widely used in medicine for drawing of blood from swollen area and also to reduce fluid pressures in damaged tissues of animals.
 - Recently, *Leeches* have been used as therapy in Veterinary Medicine to treat disease of domestic animals.
 - The most common indication of *Leech* therapy are including diseases which concerned with inflammation, hip and elbow dysplasia, disease of tendons, ligament and fascia, disease of vertebrae and treatment of scar, and also other indications of *Leech* therapy in domestic animals
- I. The researcher should have to undertake further study on therapeutic importance of *Leech* in domestic animals
 - II. Identification of *Leech* species and evaluation of bioactive substances in saliva of *Leech* should have to be studied in Ethiopia.
 - III. The species of *Leech* which are not present in Ethiopia should be imported.
 - IV. Therapeutic importance of *Leech* should have to be applied in Ethiopia.
 - V. Animal health sector and government ought to install drugs and other infrastructure which needs for treatment and control of *Leech* infestation.

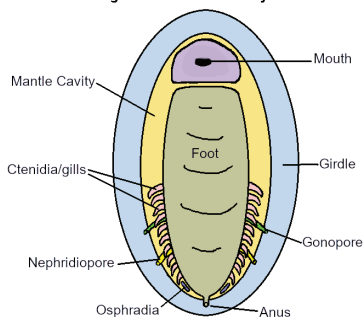
4.11. BIOLOGICAL SIGNIFICANCE LIMULUS



- The horseshoe crab (*Limulus polyphemus*), is **the oldest living fossil in Maryland**. Horseshoe Crabs evolved much earlier than humans or the Chesapeake Bay.
- Horseshoe crabs are an important part of the ecology of coastal communities. Their eggs are **the major food source for shorebirds migrating north**, including the federally-threatened red knot.
- Horseshoes are a vital part of the ecology of coastal communities. Their eggs are a major food source for many fish and northward migrating shorebirds, including the federally threatened red knot. These shorebirds have evolved to time their migrations to coincide with horseshoe crab spawning.
- Adult horseshoes serve as prey for sea turtles, alligators, conchs and sharks
- Their unique blue blood holds a compound that's used by pharmaceutical companies to ensure that their intravenous drugs are free of potentially harmful bacteria. After extraction, the animals are released back into the wild.
- Horseshoe crabs are known for their large nesting aggregations, or groups, on beaches. In Florida, they can nest year-round, with peak spawning occurring in the spring and fall.
- Horseshoe crabs are an ancient and unique species that benefit humans as well as the natural world. These creatures are unlike anything on the planet...

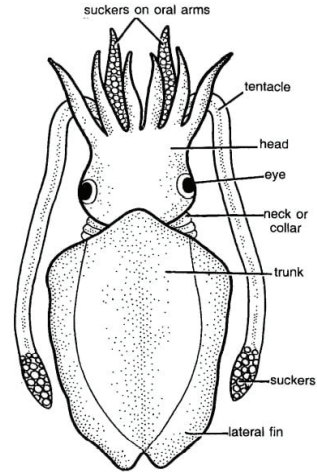
4.12. BIOLOGICAL SIGNIFICANCE CHITON

Generalized Diagram of Chiton Anatomy as seen from Below



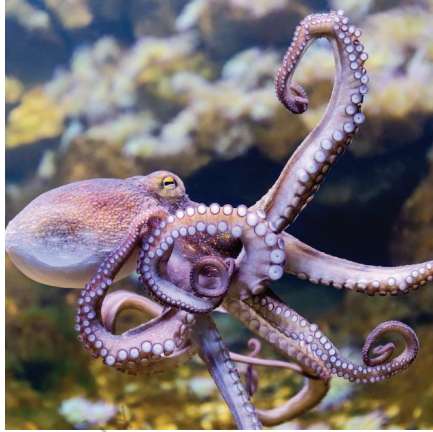
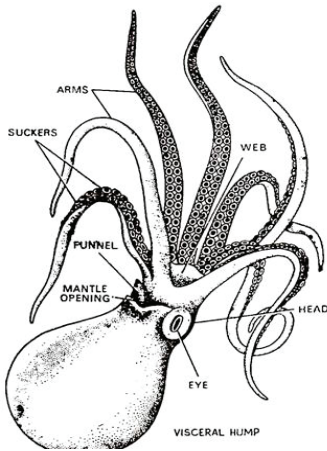
- A chiton **secretes a mucus that swells on contact with water**. This tough mantle completely covers the chiton's eight transverse shell plates, and prevents foreign organisms from attaching to the spines of the girdle.
- Even though chitons are important for their **role as primary consumers of marine plants**, many chitons feed predominantly on animals, for example, grazing on encrusting colonial animals in the low intertidal or on sponges or foraminifera in the deep sea or associated with deep sunken wood or even deep-sea hydrothermal ...
- They are also sometimes known as sea cradles or "coat-of-mail shells", or more formally as loricates, polyplacophorans, and occasionally as polyplacophores.
- Chitons have a shell composed of eight separate shell plates or valves.
- The shell plates are encircled by a skirt known as a girdle

4.13.BIOLOGICAL SIGNIFICANCE SEPIA



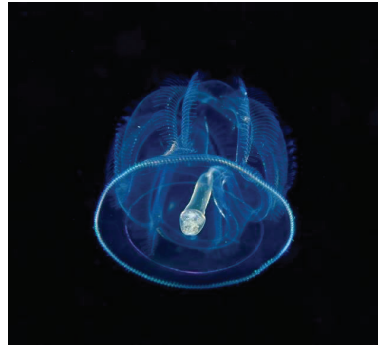
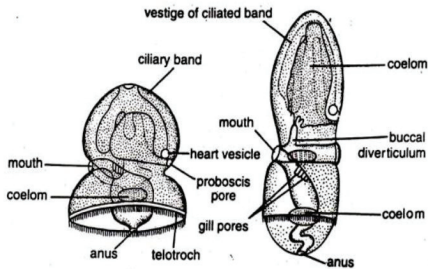
- Sepia species feed mainly on crustaceans, small fishes, and each other. Their main enemies are large aquatic animals.
- Cuttlefish are used by humans as food, as a source of ink, and for the cuttlebone, a dietary supplement providing calcium for cage birds.
- Cuttlefish are used by humans as food, as a source of ink, and for the cuttlebone, a dietary supplement providing calcium for cage birds.
- Cuttlefish have an amazing adaptation for camouflage (changing color and shape). Their cells can expand and contract to change into a multitude of different colors to match their environment.
- They can also change the texture of their skin to match the ocean floor. ... -Cuttlefish have one bone called the cuttlebone.
- Cuttlefish is especially popular in Italy, where it is used in *Risotto al Nero di Seppia* (literally *black cuttlefish rice*).
- Spanish cuisine, especially that of the coastal regions, uses cuttlefish and squid ink for the marine flavor and smoothness it provides; it is included in dishes such as rice, pasta, and fish stews.

4.13. BIOLOGICAL SIGNIFICANCE OCTOBUS



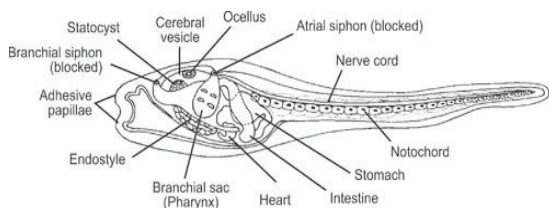
- These octopuses are foraging predators on coral reefs of the Caribbean Sea. ... This species is eaten by most large bony fishes and sharks that live on the reef.
- In order to escape predation, Caribbean reef octopuses **can eject a cloud of dark ink** toward an oncoming predator.
- Octopus is an **excellent source of omega-3 fatty acids**, "good fats" linked to a range of heart-healthy benefits. Omega-3s can lower your blood pressure and slow the buildup of plaque in your arteries, reducing stress on the heart.
- Octopuses play a key predatory role in shaping communities that is unmatched by any other invertebrate. Additionally, their "live fast, die young" life history strategy means their **populations respond quickly and dramatically to changes in the environment**, which suggests they could be an indicator of ecosystem change.
- Expanding or contracting the cells, **octopuses can quickly change color and ultimately blend into their environment**
- By imitating these toxic animals, mimic octopuses can protect themselves from predators while vulnerable in the open ocean.

4.14. BIOLOGICAL SIGNIFICANCE TORNARIA LARVA



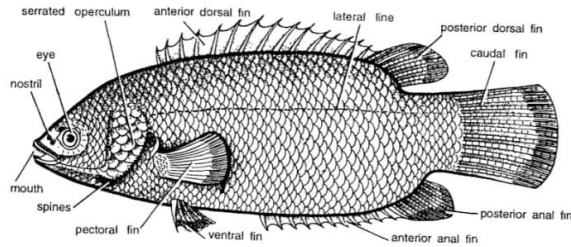
- A tornaria larva is the planktonic larva of some species of Hemichordata.
- It is very similar in appearance to the bipinnaria larvae of starfishes, with convoluted bands of cilia running around the body.
- A tornaria is the planktonic larva of some species of Hemichordata such as the acorn worms. It is an oval-shaped, transparent larva.
- The diameter of the body is about 3mm. It has an apical plate, which is a thickened region provided by a tuft of cilia and a pair of eyespots.
- The larva has a complete alimentary canal. The ciliary band stretches throughout the anterior and posterior region, and also the posted region.
- Balanoglossus is less sensitive in response to external stimuli. I take sand- containing diatoms, protozoa and many other microorganisms, the tornaria larva of Balanoglossus is a free-swimming planktonic form which exploits the natural resources of the sea
- Balanoglossus emits a peculiar odor resembling that of iodoform.
- This is regarded as a protective device, it is a sluggish animal and proboscis is the most active part of the body.

4.15. BIOLOGICAL SIGNIFICANCE- ASCIDIAN TADEPOLE LARVA



- Ascidian Larva. Ascidian larvae exhibit the **simplest chordate body plan**, with a head and a tail containing a notochord, three rows of muscle cells, and a hollow dorsal nerve.
- The adult tunicates reflect the primitive sessile, marine and filter feeding condition of the ancestral chordates.
- Retrogressive metamorphosis is characteristic of ascidians (Urochordata or Tunicata). It is given this name because here a **progressive, active and alert larva metamorphoses into a retrograde and sedentary adult**.
- Most of these characters are lost or become degenerate as the active larva metamorphosis's.
- Retrogressive metamorphosis is a type of metamorphosis seen in **Herdmania (and other urochordates)**.
- It involves transformation of an active, free swimming larva with advanced characters into sedentary and simple adult.

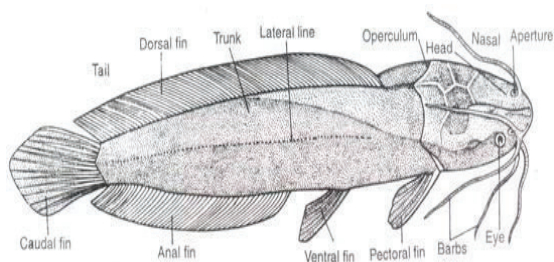
4.16. BIOLOGICAL SIGNIFICANCE-ANABAS SCANDENS



- Anabas is able to walk on land by spines in search of earthworms. The crows and kites attack them and take their bodies over trees, and thus the fish is called as climbing perch,
- Anabas is so much dependent on atmospheric oxygen that it is asphyxiated if kept in water with no access to air. Fish can live out of water for a long period.
- **Anabas is able to walk on land by spines in search of earthworms.** The crows and kites attack them and take their bodies over trees, and thus the fish is called as climbing perch as the fish might have climbed trees.
- This species is important as a **food fish** in certain areas of South Asia and in Southeast Asia, where its ability to survive out of the water for extended periods of time, provided it is kept moist, improves its marketability.

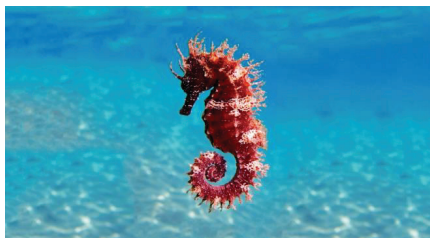
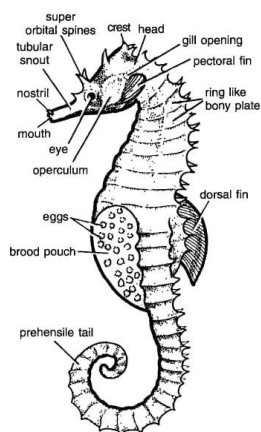


4.18. BIOLOGICAL SIGNIFICANCE CLARIAS BATRACHUS



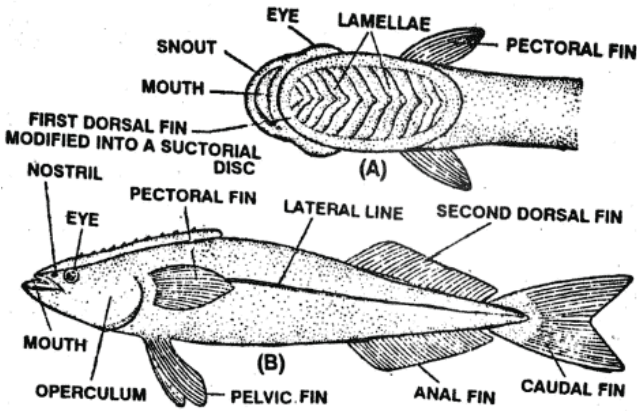
- Walking catfish put few native species at risk, as an array of wading birds, such as herons and egrets, are natural predators. However, they often feed in aquaculture ponds, "walking" from fish farm to fish farm, preying on the fish there and causing millions of dollars in damage.
- Its ability to 'walk' over land aided by its pectoral fins, combined with high fecundity and voracious feeding habits have led to this species becoming both a threat to native fish species and an invader of fish farms.
- In the wild, this creature is omnivorous; it feeds on smaller fish, molluscs, and other invertebrates, as well as detritus and aquatic weeds.
- It is a voracious eater that consumes food supplies rapidly, so it is considered harmful when invasive.
- Not true *C. batrachus*, but *C. magur* is eaten in West Bengal and is considered good during illness, particularly for body weakness. It is prepared in a light curry sauce with coriander powder and cinnamon powder. It reportedly is fed to children to develop body strength.
- In Karnataka, *C. magur* is called *murgodu*. In coastal Karnataka, it is called *mugudu* (Tulu), and is considered a delicacy. Neither the Thai nor the Indian populations are likely to be *C. batrachus*, however.

4.19. BIOLOGICAL SIGNIFICANCE HIPPOCAMPUS



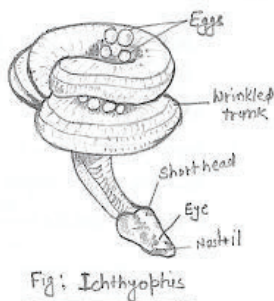
- Seahorses are truly unique, and not just because of their unusual equine shape. Unlike most other fish, they are monogamous and mate for life. Rarer still, they are among the only animal species on Earth in which the male bears the unborn young.
- Seahorses, a syngnathidae fish, are one of the important candidate organisms which have been used in Chinese traditional medicine from time immemorial.
- It is believed that seahorses have the potential to cure infertility, baldness, asthma and arthritis. The symbolic meaning of Seahorse is quite intricate and diverse as this little creature itself is full of surprises.
- Seahorses are also consumed by Indonesians, central Filipinos, and many other ethnic groups
- The preferred large, pale, and smooth seahorses has been offset by the shift towards prepackaged preparations, which makes it possible for TCM merchants to sell previously unused, or otherwise undesirable juvenile, spiny, and dark-colored animals

4.20. BIOLOGICAL SIGNIFICANCE ECHENEIS



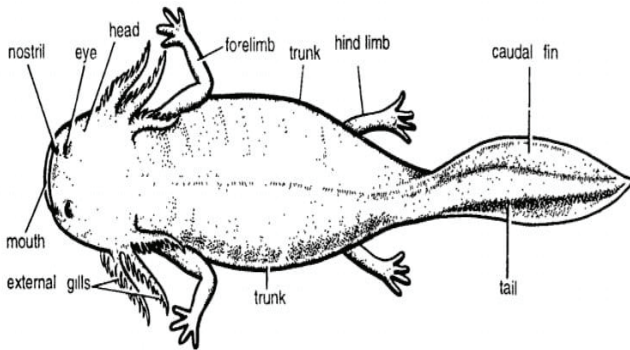
- These are very recognizable fish because of their highly modified dorsal fin that is an oval shaped sucking disc.
- They are as long as 43 inches, and slender, with lower jaws that extend much further than upper.
- They attach themselves to sharks, turtles, whales, large bony fish, and sometimes ships or human divers, with the goal of eating parasites and food scraps from the host.
- Humans have been known to use these sharksuckers in a unique method to catch larger fish and turtles.
- Although the sharksucker is of minor commercial importance, it is considered a gamefish by recreational fishers. It is also captured and displayed in public aquaria facilities.
- An interesting aspect of the importance of this unique fish to humans is that it is used by natives to catch other fish.
- A fishing line is tied to the caudal peduncle of the sharksucker, which is then released, into the deep sea. When the sharksucker attaches to another fish, the sharksucker and its host is then hauled in to land or to the boat by the fisher.

4.21. BIOLOGICAL SIGNIFICANCE ICHTHYOPHIS



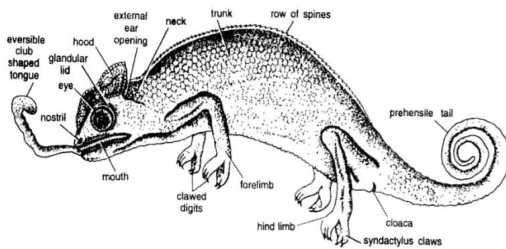
- Ichthyophis is a genus of caecilians (limbless amphibians, sometimes called the Asian caecilians) found in Southeast Asia, the southern Philippines, and the western Indo-Australian Archipelago. In Sri Lanka, three species occur.
- Ichthyophis malabarensis too fertilization is internal. either copulation or fertilization.
- Basal caecilians such as Ichthyophis go through a metamorphosis in which aquatic larva transition into fossorial adults, which involves a loss of the lateral line.
- Thus, most caecilians do not undergo an anuran-like metamorphosis.
- It's mostly assumed that caecilians are generalist predators of soil-dwelling invertebrates such as earthworms, ants and termites – animals termed 'soil ecosystem engineers' by ecologists.
- Caecilians are intriguing animals. They look like worms or snakes, but they are actually amphibians.
- Scientists have known for some time that the animals are poisonous. Recent evidence suggests that they may be venomous as well.
- Animals called caecilians may have been among the first vertebrates on land to lace their bites with venom.

4.22. BIOLOGICAL SIGNIFICANCE AXOLOTL LARVA



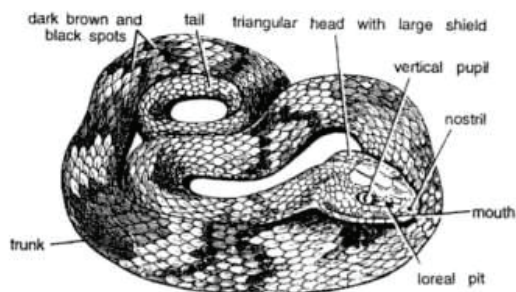
- Axolotls are used a lot in scientific research because they can regenerate most body parts, are easy to breed, and have large embryos.
- Axolotls are closely related to waterdogs, the larval stage of the closely related Tiger salamanders *Ambystoma tigrinum* and *Ambystoma mavortium*.
- The Axolotl plays an important role in Mexico's life style and survival. Before they were endangered they were a valuable food source for the people of Mexico.
- Axolotls and other salamanders around the world are becoming or are extinct due to factors such as pesticides, predators and habitat destruction.
- These creatures have a rare ability. Because of their ability to grow up and leave the water, experts classify axolotls as amphibians.
- One of the Axolotls unique adaptations is the capability of regenerating lost body parts.
- If an axolotls spends long periods of time in shallow water, their gills may be absorbed and they can begin using their lungs on land.
- However, if Axolotls are in deep water, they will keep their gills.

4.23. BIOLOGICAL SIGNIFICANCE CHAMAELEON



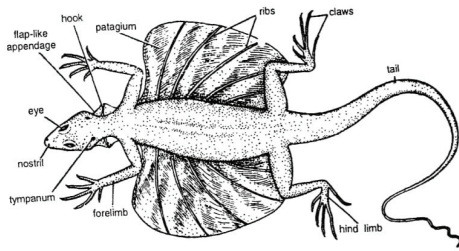
- The chameleon spirit animal has a shining personality, always displaying courage and boldness with every human contact.
- The chameleon meaning also speaks about having a clearer perception and a stronger intuition.
- The chameleon symbolism represents a period of changes. You need to learn to adjust to your environment and blend in to create harmony.
- The chameleon spirit animal symbolizes sensitivity and clairvoyance.
- It encourages you to have patience, to gain insight, to always be curious, and to be resourceful.
- Some benefits of having a chameleon as a pet include the fact that they do not need much direct interaction from their owners and have the unique ability to change colors.
- Chameleons themselves are generally odorless but their enclosure can give off smells if things aren't tended to regularly and feeder insects not regularly cleaned. Chameleons are solitary animals.
- Forced handling or unwanted handling can cause hissing and biting. A chameleons bite is painful, however, not toxic or harmful to humans.
- Handling can cause chameleons to have chronic low-level stress, which leads to poor health.

4.24. BIOLOGICAL SIGNIFICANCE VIPERARUSSELLI(RUSSEL VIPER)



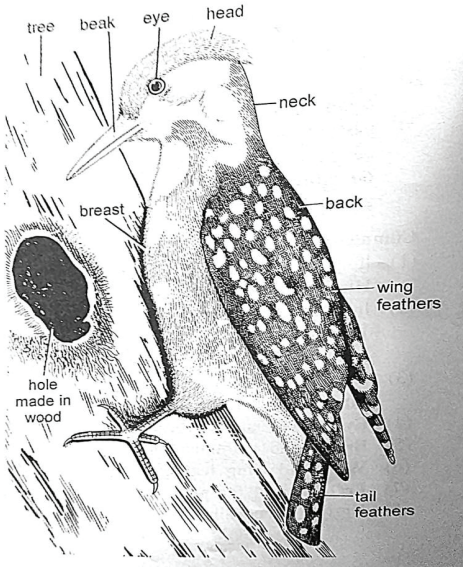
- Russell's viper bite is an occupational hazard of rice farmers throughout its geographical range. Defibrination, spontaneous haemorrhage, shock and renal failure develop with frightening rapidity.
- In several countries, Russell's viper bite is the commonest cause of acute renal failure.
- Russell's viper (*Daboia russelii*) is one of the most dangerous snakes in the world. It is deadly venomous and kills many people each year.
- Russell's viper venom (RVV) was known to clot blood many years ago. It was widely used as a styptic to clot minor wounds when razor blades were more commonly used for shaving (e.g. "Stypven", Burroughs-Wellcome Pharma).
- Snake venom contains several neurotoxic, cardiotoxic, cytotoxic, nerve growth factor, lectins, disintegrins, haemorrhagins and many other different enzymes. These proteins not only inflict death to animals and humans, but can also be used for the treatment of thrombosis, arthritis, cancer and many other diseases.
- Distilled venom from these animals has been effective when treating people with infections of various microorganisms that can cause pneumonia, staphylococcus, urinary tract infections, and more.

4.25. BIOLOGICAL SIGNIFICANCE DRACOVOLANS



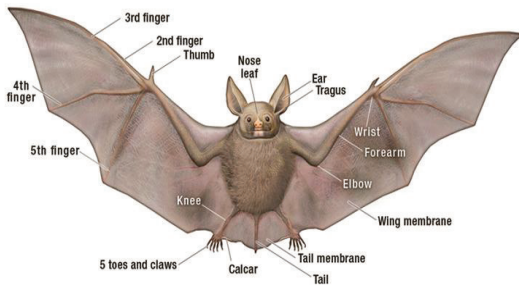
- *Draco volans*, also commonly known as the common flying dragon, is a species of lizard in the family Agamidae.
- The species is endemic to Southeast Asia. Like other members of genus *Draco*, this species has the ability to glide using winglike lateral extensions of skin called patagia.
- *Draco* is a genus of agamid lizards that are also known as flying lizards, flying dragons or gliding lizards.
- These lizards are capable of gliding flight via membranes that may be extended to create wings (patagia), formed by an enlarged set of ribs.
- Flying lizards are skittish and need a huge enclosure full of vegetation and even small trees. Indeed, there are places where it is illegal to own one of these skittish lizards.
- It's thought that the lizard's chief predators are arboreal (tree-living) snakes, large birds, and monitor lizards.
- Despite the presence of their predators, most flying dragons are very successful in their habitat.
- *Draco volans*, also commonly known as the common flying dragon, is a species of lizard in the family Agamidae.

4.25. BIOLOGICAL SIGNIFICANCE DINOPIUM(WOOD PECKER)



- ***Inopium*** is a genus of birds in the woodpecker family Picidae. The species are found in South and Southeast Asia.
- The genus was introduced by the French polymath Constantine Samuel Rafinesque in 1814 to accommodate the common flameback (*Dinopium javanense*).
- Three-toed, small-billed, golden-backed woodpecker found primarily in wet tropical forests in two disjunct regions, the Western Ghats in India and Southeast Asia.
- Separated from Greater Flameback by a much smaller bill, a black nape, and a black undivided horizontal stripe below the cheek.
- Separated from Black-rumped Flameback by a red rump and a white throat.
- This species is a regular participant in mixed-species foraging flocks, often detected by its loud, high-pitched trill similar to that of Greater Flameback.
- Best told from the nearly identical Himalayan Flameback by voice.

4.27. BIOLOGICAL SIGNIFICANCE BAT



- Bat conservation is important not only for biodiversity but also because these flying mammals provide ecological services essential for humans. I
- Bats are very useful for the control of insect populations and specifically pests to agriculture. Their diet very often includes Lepidoptera, a large order with several species very harmful to many plants of great economic interest.
- Insect-eating bats are great for keeping bugs away from crops, as well as the places where the bats roost.
- The Brazilian free-tailed bat has been recognised as an important “pest management service” in cotton farming. Because bats eat so many insects in some regions, they can also reduce the need for pesticide sprays.
- Natural enemies of insect pests, also known as biological control agents, include predators, parasitoids, pathogens, and competitors.
- Fruit-eating bats play important roles in distributing seeds to maintain plants and forests. These species of bats, often called "flying foxes" because of their larger body size and big eyes, live in tropical and subtropical areas of the Old World (Africa, Asia and Australia). Bats are of major environmental significance due to being important indicators of how well an ecosystem is doing.

COMMENT ON STRUCTURE/SKELETON/PALATE/DENTITION

5.1.Sponge-spicules

5.2.Sponge-gemmule

5.3.Taenia-scolex

5.4.Neantes-parapodium

5.5.Penaeus-petasma

5.6.Scorpion-book lung

5.7.Starfish-pedicellaria

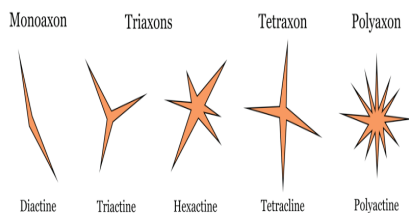
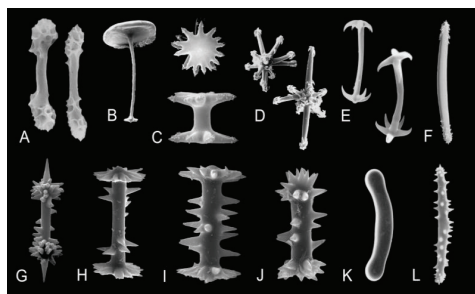
5.8.Rana-pectoralgirdle

5.9.Rana-pelvic girdle

5.10.Pigeon-palate

5.11.Rabbit-dentition

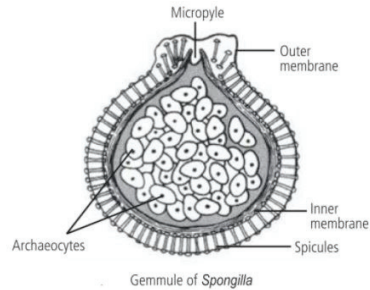
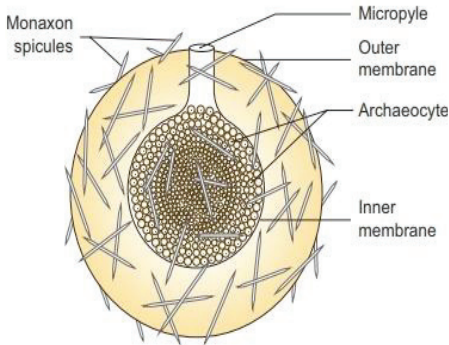
5.1. SPONGE-SPICULES



COMMENTS:-

- Sponge body wall is supported by various minute, crystalline and bodies called as spicules, which are secreted by special mesenchymal cells as Scleroblasts.
- Spicules are of two types.
 - (i) Megascleres or supporting skeleton,
 - (ii) Microscleres small and non-supporting.
- Monaxon consists of a single axis, straight or curved. They may be styles, rhabds and tylots. Tetraaxon consists of four rays. It also includes triradiate or triactinal spicules. Polyaxon having several equal rays.
- Amphidiscs spicules are found in fresh water sponges. In this type, the rhabdom contains disks at both ends.
 - The arrangement of different types of spicules could be seen in Sycon.
- Microscleres are found throughout the mesenchyme and include spires and asters.
- Spicules support and protect the body.
- They are helpful in identification,

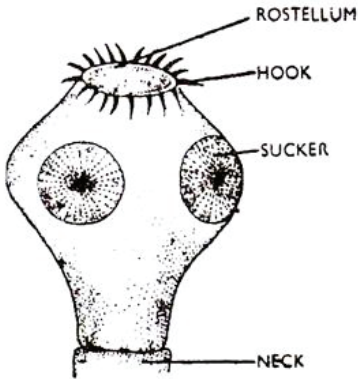
5.2. SPONGE-GEMMULE



COMMENTS:-

- Gemmule is Asexual reproductive bodies forming a part of regular life cycle.
- These are endogenous buds which are diagnostics of Porifera, especially of freshwater and a few marine sponges.
- Gemmulation or endogenous budding is a peculiar mode of reproduction under unfavourable condition such as excessive cold or drought.
- Gemmule is resistant to external factors such as freezing and drying. Gemmule of fresh water sponge can be kept for 2 years.
- They hatch at a temperature of 13-21 C in about 3 days.
- After hatching, a gemmule gives rise to sponge.

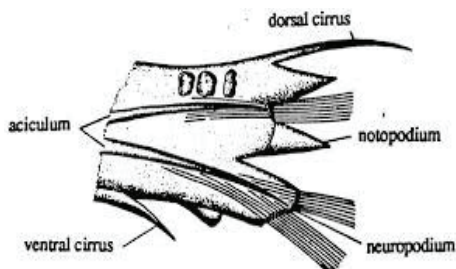
5.3. TAENIA-SCOLEX



COMMENTS:-

- In tapeworm the the science is the organ of attachment.
- It hears **at** its apex a small prominent known as rostellum.
- Rosetellum is protrussible.
- At the base of rosetellum, there is a double circlet of eversible chitinous hooks.
- Four circular unarmed sucker or present.
- Hooks and suckers serve to fix the animal to the intestine other human host.

5.4. NEANTHES-PARAPODIUM



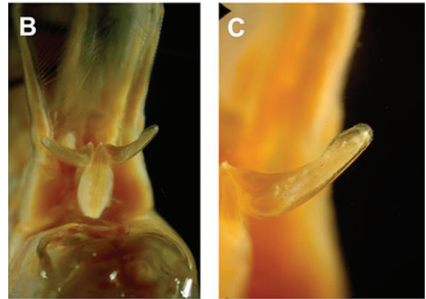
COMMENTS:

- Parapodia are predominantly found in annelids, where they are paired, unjointed lateral outgrowths that bear the chaetae.
- In several groups of sea snails and sea slugs, 'parapodium' refers to lateral fleshy protrusions

Parapodia in Neanthes:

- Parapodia are fleshy, paddle like locomotory appendages, arising from the lateral side of the body segments in polychaetes.
- In Neanthes, all body segments except the first and last segments bear on either lateral side the parapodia.
- The neuropodium has one ventral cirrus and one ligule.
- Most species of polychaete annelids have paired, fleshy parapodia which are segmentally arranged along the body axis.
- Parapodia vary greatly in size and form, reflecting their variety of functions, such as gas exchange, anchorage, protection and locomotion.

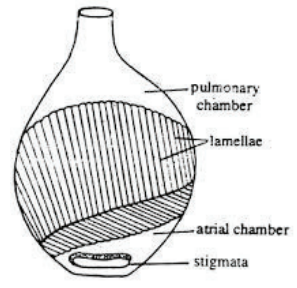
5.5. PENAEUS-PETASMA



COMMENTS:

- The petasma is copulatory organ present only in the male penaeus.
- It is a membranous structure formed by the fusion of the endopodites of the first abdominal appendages.
- The male has a pair of tubular testes located in the thorax on the either side of the middle line.
- The two testes are fused together anteriorly each testes has many finger like outgrowth called caecal diverticula.
- Posteriorly the testes leads into air vasdeferens .
- The terminal end of the vasdeferens becomes dilated into a ejaculatory bulb.
- The ejaculatory bulb opens to the outside by the male genital pore at the base of the last walking legs.

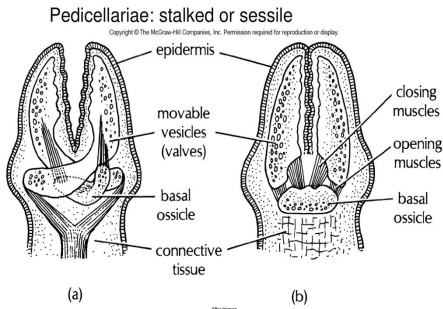
5.6. SCORPION-BOOK LUNG



COMMENTS:

- Living scorpions have four pairs of book-lungs, each pair situated above a sternite on the ventral surface of the mesosoma (anterior abdomen) and each book-lung opening to the outside through a stigma which perforates the sternite.
- A rudimentary type of lung typically found in arachnids, such as scorpions and spiders, and in horseshoe crabs. The name describes the structure, as it contains many thin folds of membrane resembling the “leaves” of a book.
- A book lung is a type of respiration organ used for atmospheric gas exchange that is present in many arachnids, such as scorpions and spiders.
- Book lung, form of respiratory organ found in certain air-breathing arachnid arthropods (scorpions and some spiders).
- Each book lung consists of a series of thin plates that are highly vascular (i.e., richly supplied with blood) and is arranged in relation to each other like the pages of a book.

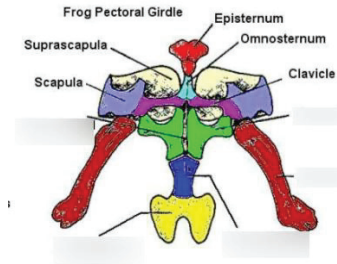
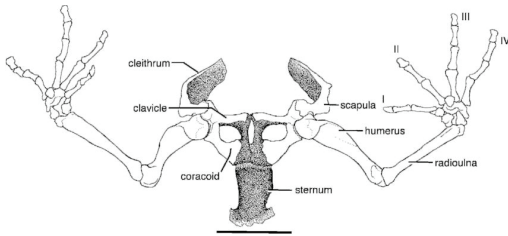
5.7. STARFISH-PEDICELLARIA



COMMENTS:

- A pedicellaria (plural: pedicellariae) is a small wrench- or claw-shaped appendage with movable jaws, called valves, commonly found on echinoderms (phylum Echinodermata), particularly in sea stars (class Asteroidea) and sea urchins (class Echinoidea).
- The presumed functions of *E. cordatum* pedicellariae are (1) cleaning of the body surface and ciliary structures (trifoliate), (2) protection against sedimenting particles (tridactyles), and (3) defense of the peribuccal area against potential small predators (globiferous pedicellariae).
- Pedicellariae attach and articulate to small granules on the surface of the test.
- In life the pedicellariae are in constant motion. The valves that form the head can open and close and the primary function of the pedicellariae is to keep the surface of the test free of debris and small parasites.

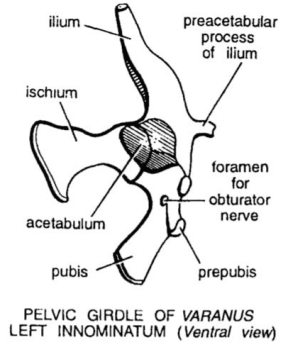
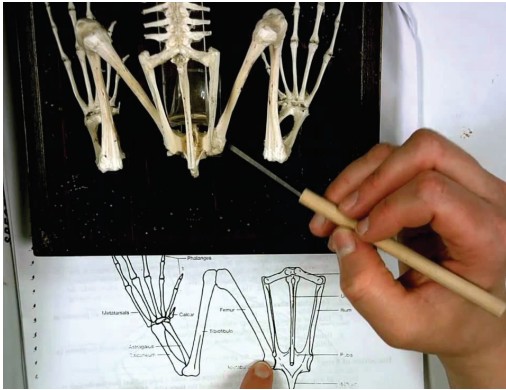
5.8. RANA-PECTORALGIRDLE



COMMENTS:

- A pectoral girdle, also referred to as the shoulder girdle, connects your upper limbs to the bones along the axis of your body. You have two pectoral girdles in your body.
- The pectoral girdle consists of the two bones that make up your shoulder: clavicle, or collarbone.
- The pectoral girdle or shoulder girdle is present in the thoracic region and provides attachment to the forelimbs and their muscles. It also protects the inner softer parts of the thorax.
- It consists of two similar halves united mid-ventrally and separated dorsally.
- The shoulder, or pectoral, girdle is composed of the clavicles (collarbones) and the scapulae (shoulder blades).

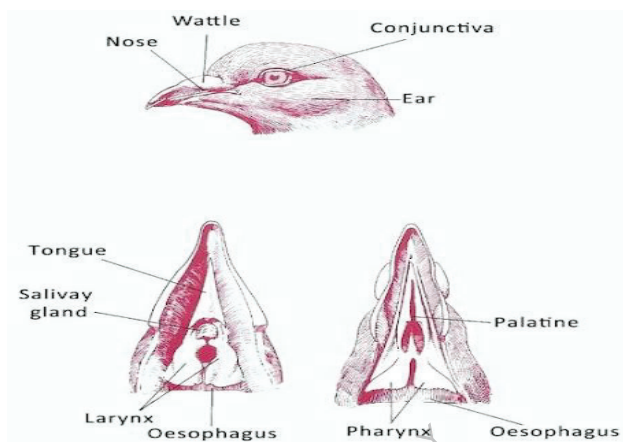
5.9. RANA-PELVIC GIRDLE



COMMENTS:

- The pelvic girdle has two equal halves which are known as 'ossa innominata'. Each as innominatum is formed by three bones.
- They are the dorsal bone ilium, the ventral bone-ischium and the antero-ventral bone pubis. The pelvic girdle has a depression at the junction of the three bones.
- The pectoral girdle has been suggested to function as a damper and absorber of the landing impact (Nauwelaerts and Aerts 2006;Emerson 1983).
- The frog pelvis is unique, with flared sacral extensions and elongate iliac rods that support the ischium posteriorly, all surrounding the urostyle.

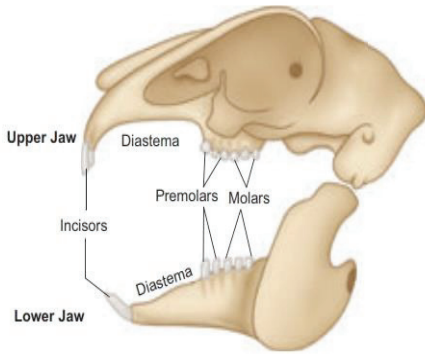
5.10. PIGEON-PALATE



COMMENTS:

- Palate, in vertebrate anatomy, the roof of the mouth, separating the oral and nasal cavities.
- The roof of the mouth is made of an incomplete hard palate. Nose's internal part is located between the palatal bones and the vomer, caudal to the palatal process of the maxillary bone.
- The soft palate is composed of muscle and connective tissue, which give it both mobility and support.
- This palate is very flexible. When elevated for swallowing and sucking, it completely blocks and separates the nasal cavity and nasal portion of the pharynx from the mouth and the oral part of the pharynx. While elevated, the soft palate creates a vacuum in the oral cavity, which keeps food out of the respiratory tract.
- The first well-developed palates are found in the reptiles, although only in the form of a hard partition.
- Palates similar to those in humans occur only in birds and some mammals. In a few whales the mucous membrane forms toughened plates known as baleen, or whalebone.

5.11.RABBIT-DENTITION



COMMENTS:

- Rabbits have incisor and cheek teeth. The cheek teeth include both premolars and molars.
- Rabbits do not have canine teeth as in cats, dogs, ferrets and hedgehogs. Rabbits have a diphyodont dentition since they have deciduous (primary) and secondary (adult) teeth.
- The dental formula for a rabbit is $12/1$, $C0/0$, $P3/2$, and $M3/3$, for a total of 28 teeth. Rabbit teeth are cylindrical and have a natural curve as they grow.
- The occlusal surface of the maxillary cheek teeth curves buccally, and the occlusal surface of the mandibular cheek teeth curves lingually.
- The dentition of rabbit is called heterodont, as they have different types of teeth (Incisors, premolars and molars).

GENERAL LABORATORY SAFETY PRECAUTION

1. Determine the location of fire extinguishers, chemical safety showers and eye washers, chemical spill kits and alternative exit routes for lab evacuation.
2. Smoking, eating or drinking in the lab room is totally prohibited.
3. Wear lab aprons when working with chemicals, hot material or preserved specimens.
4. Wear safety goggles when using dangerous chemicals, hot liquids or burners.
5. Any chemicals spilled on the hands or other parts of the skin should be washed off immediately with a plenty of running water.
6. If you have an open skin wound, be sure that it is covered with a water proof bandage.
7. Never work alone in the laboratory.
8. Keep your work area clean and dry.
9. Turn off all electrical equipment, water and gas when it is not in use, especially at the end of the laboratory period.
10. Tie back long hair.
11. Report all chemicals spills or fluids to your instructor immediately for proper clean up.

Special precautions for working with heat or fire:

1. Never leave a lighted Bunsen burner or hot object unattended. When an object is removed from the heat & left to cool, it should be placed where it is shielded from contact.
2. Inflammable liquid bottles should not be left open, not dispensed near a naked flame, hot electric element or electric motor.
3. Use test tube holders to handle hot laboratory equipments.
4. To heating something in a container such as a test tube, always point the open end of the container away from yourself & others.
5. Use only Pyrex glassware's for heating.
6. Allow hot materials to cool before moving them from your lab station.
7. Make sure that Bunsen burner hoses fit tightly.

Special precautions for working with chemicals

1. Never taste or touch substances in the laboratory without specific instructions.
2. Never smell substances in the laboratory without specific instructions.
3. Use materials only from containers that are properly labeled.
4. Wash your hand after working with chemicals.
5. Do not add water to acid. Instead, dilute the acid by adding it to water.

6. Mix heat generating chemicals slowly.

Special precautions for working with electrical equipment

1. Make sure the area under & around the electrical equipment is dry.
2. Never touch electrical equipment with wet hands.
3. Make sure the area surrounding the electrical equipment is free of flammable materials.
4. Turn off all power switches before plugging an appliance into an outlet.

Special Precaution for working with Glassware's and other laboratory equipments

1. Become familiar with the names and appearance of all the laboratory equipments you will use.
2. Never use broken or chipped glassware.
3. Make sure that all glassware's are clean before you using it.
4. Do not pick up broken glass with your bare hands. Use a pan and a brush.
5. If a Mercury thermometer breaks, do not touch the mercury.
6. Do not aim the mirror of your microscope directly at the sun. Direct sun light can damage the eyes.
7. Use care handling all sharp equipments, such as scalpels and dissecting needles.

Special precautions for working with live or preserved specimens

1. If live animals are used treat them gently. Follow instructions for their proper care.
2. Always wash your hands after working with live or preserved organisms.
3. Specimens for dissection should be properly mounted and supported. Do not try to cut a specimen while holding it in the air.
4. Do not open Petri dishes containing live cultures unless you are directed to do so.
5. Detergents (detol 5 – 10%) should be used to sterilize and clean benches, glassware and equipment.
6. Safety cabinet should be used while working with microbes.
7. Lab coats should be worn during the work in the lab.
8. Disposable items should be collected and autoclaved.

First Aid

1. **Injuries:** bleeding should be reduced using bandages; the wound should be cleaned with iodine alcohol mixture and wrapped with sterile bandage.
2. **Acid and fire burns:** body burns must be washed immediately with tap water. Eye burns must be washed using eye washer, special cream for burns can be used.
3. **Poisoning:** if any toxic chemical is swallowed, the mouth must be sensed with water, in case of acid, milk is drunk, in case of alkaline, diluted acetic acid (vinegar) can be used.
4. Skin contamination requires washing with water and removal of contaminated clothing, if the contaminant is insoluble in water remove with soap and water.

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MODEL ANIMAL HOUSE PERMISSION LETTER

To

13-04-2022

The Principal

J.K.K.Nattraja College of Pharmacy,

Natrajapuram,

Komarapalayam- 638 183

Respected Sir,

With due respect, I am penning this letter to you to seek your permission to visit Animal house on 18th April, 2022. The visit will help the Zoology students to gain knowledge about laboratory animal maintenance, handling and ethical policies etc., It is an opportunity for involve in vital part of live research. Kindly give your due concern for the same.

Thanking you,

Yours truly,

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From
Dr. S. KALAIMANI
Assistant Professor
PG and Research Department of Zoology
J. K. K. Nataraja College of Arts and Science
Komarapalayam-638 183.

Date: 13.04.2022
Place: Komarapalayam

To
THE PRINCIPAL
J. K. K. Nataraja College of Arts and Science
Komarapalayam-638 183.

Respected Madam,

Sub: Permission for Animal house visit-Reg.

I wish to bring to your kind consideration that we have planned to visit Animal house at J.K.K. Nataraja College of Pharmacy, on 18.04.2022 for the I B.Sc. Zoology students (17 students). The visit will help the Zoology students to gain knowledge about laboratory animal maintenance, handling and ethical policies etc., It is an opportunity for involve in vital part of live research. Hence, I kindly request you grant us permission to take a visit for the above-mentioned students. I herewith attach the list of students for your kind perusal.

Encl.: List of Students

Yours faithfully,

(Dr. S. KALAIMANI)

J.K.K. Nataraja College of Arts and Science, Komarapalayam- 638 183

PG and Research Department of Zoology

Students' Name list- I. B.Sc. Zoology (2021-2022)

1. Abinaya.K
2. Arthi.M
3. Devi.S
4. Karthika.M
5. Kavipriya.N
6. Myla.R
7. Narchonai.E
8. Nirmaladevi.M
9. Pavithra.K
10. Pavithra.M
11. Reka.V
12. Sathishkumar.M
13. Snega.Y
14. Swetha.R
15. Sudha.S
16. Tamilselvi.S
17. Thenmozhi .S

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